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Physics informed machine learning based predictive control for intelligent operation of edge datacenters

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HIGHLIGHTS

- Propose a PIML-MPC approach for improved prediction accuracy and control robustness.
- Embed tailored physical equations into the NN to ensure physical hard conservation.
- PIML-MPC surpasses traditional fixed control across all operating environments.
- PIML-MPC outperforms three widely used time-series and ML-based MPC methods.
- Provide in-depth, AI-driven insights into intelligent control of cooling systems.

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ABSTRACT

Edge data centers (EDCs) are critical for supporting latency-sensitive applications; however, their high energy consumption, especially in cooling systems, presents significant sustainability challenges that are further compounded by recent advancements in Artificial Intelligence (AI). In recent years, integrating AI with Model Predictive Control (MPC) has shown promise in reducing cooling energy consumption while maintaining operational safety, but most AI models remain impractical for real-world deployment. This limitation arises from their inherent black-box nature and heavy dependence on the quantity and quality of training data, which hampers their ability to handle unseen conditions effectively. To address these challenges, we propose a physics-informed machine learning-based MPC (PIML-MPC) framework that integrates physical principles into neural networks to improve prediction accuracy. By embedding established physical knowledge, our approach enhances model generalization and reduces its dependence on extensive training data. We validated the framework using a simulation model derived from real data center measurements. Experimental results show that the PIML-MPC strategy reduces energy consumption by 12.6 % compared to a baseline, without compromising thermal regulation. Under unseen high heat load conditions, the framework further reduces energy consumption by 9.1 % while maintaining thermal performance, demonstrating robust generalization. Moreover, across ten randomized trials, PIML-MPC consistently outperforms three advanced time-series and machine-learning-based MPC schemes—achieving superior energy efficiency, tighter thermal control, and reduced sensitivity to stochastic variations. This work paves the way for the practical adoption of AI-driven MPC in data center cooling by substantially improving the generalization performance of AI controllers.

1. Introduction

1.1. Background

Data centers (DCs) are fundamental to the growth of the information and communications technology (ICT) industry, accounting for approximately 2 % of global power generation—a share that continues to rise with recent advancements in Artificial Intelligence [1–4]. In particular, the rapid expansion of digital services, cloud computing, the Internet of Things (IoT), and more recently, edge AI applications have driven the development of edge data centers (EDCs). EDCs bring computing resources closer to end-users, reducing latency and enhancing performance. Unlike traditional centralized data centers, EDCs are geographically distributed to support real-time processing for critical applications such as autonomous vehicles, smart cities, and

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Nomenclature

Abbreviations

DC	Data Center
EDC	Edge Data Center
ITE	Information Technology Equipment
DCIM	Data Center Infrastructure Management
MPC	Model Prediction Control
AI	Artificial Intelligence
ML	Machine Learning
PINN	Physics-informed Neural Network
PIML	Physics-informed Machine Learning
APCNN	Adaptive Physically Consistent Neural Network
MSE	Mean Squared Error
HEX	Heat Exchanger
AHU	Air Handling Unit
CDU	Coolant Distribution Unit
TVR	Temperature Violation Rate
CSE	Cooling System Efficiency
T	Temperature
P	Power
m	Mass Flowrate
D	Pump/Fan Setpoint

Superscript

k	Timestep

Subscripts

i	Data Index
ct	Cooling Tower
cdwp	Condensed Water Pump
ch	Chiller
chwp	Chilled Water Pump

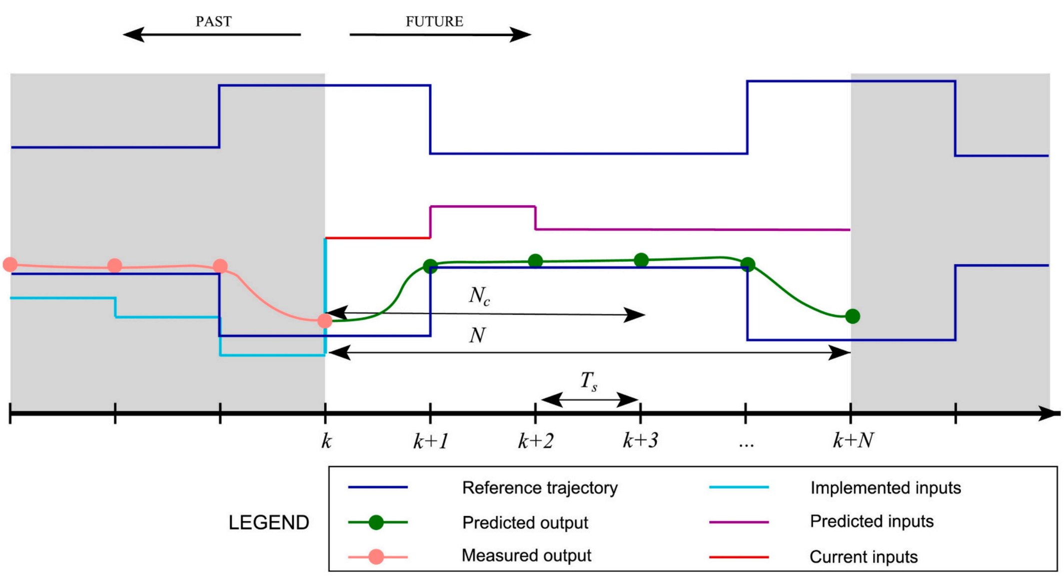

Fig. 1. Characteristic features and illustrated behavior of MPC [17].

industrial automation. However, this decentralized infrastructure introduces new challenges, particularly in terms of energy consumption and thermal management. DC energy usage is primarily driven by two components: ICT equipment and cooling systems that maintain optimal operating temperatures. Typically, DC cooling systems rely heavily on Air Handling Units (AHUs), chillers, pumps, and cooling towers, all of which are highly energy intensive. In many cases, the energy consumed by these cooling systems rivals or even exceeds that used by the ICT equipment itself [5,6].

In response to growing energy demands, various strategies have been proposed to enhance DC cooling efficiency. These include free cooling, liquid cooling, airflow management [7–9], renewable energy utilization, chips thermal management [10–12] and the adoption of intelligent control methods [13,14]. Intelligent control strategies have attracted significant attention because conventional DC cooling systems are typically configured with static settings designed to accommodate peak Information Technology Equipment (ITE) loads [15]. Such fixed settings often lead to inefficiencies: overcooling results in unnecessary energy consumption, while undercooling may compromise ITE performance and safety. In real-world operations, cooling systems are generally optimized for peak load conditions, which results in slow response times and excessive energy usage during periods of lower demand. This lack of adaptability underscores the need for dynamic, intelligent cooling strategies capable of optimizing energy use while maintaining optimal thermal conditions. Ideally, cooling parameters should adjust in response to real-time fluctuations in key variables such as ITE workload and outdoor temperature to achieve optimal cooling efficiency [16].

To overcome these challenges, advanced control strategies are essential for dynamically regulating cooling system operations and optimizing performance under continuously changing conditions. model Predictive Control (MPC) has emerged as a promising solution for DC optimization and is widely explored in industrial applications. MPC improves system performance by predicting future states and dynamically adjusting control inputs based on real-time data and predefined constraints. This predictive approach not only enhances transparency and interpretability in decision-making but also ensures more predictable and stable performance compared to traditional control methods. As shown in Fig. 1, MPC leverages predictive modeling to anticipate future system behaviors and compute control actions over a specified horizon, thereby enabling data centers to effectively address key operational objectives—including energy efficiency, thermal management, and workload balancing—while adhering to equipment operation limits

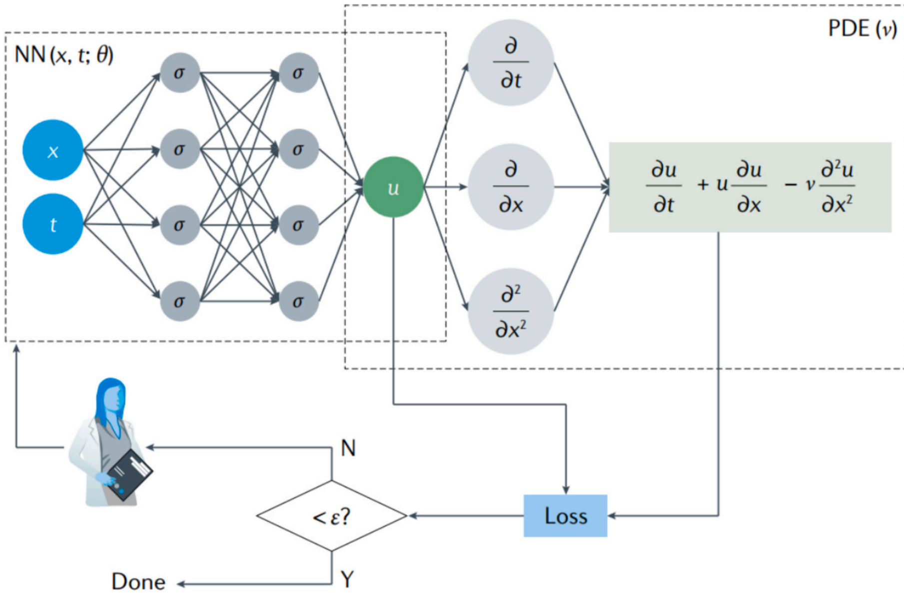

Fig. 2. Schematic of a PINN. The network is trained on measured data and PDE residual points, with a loss function combining both [24].

and service level agreements [17–21].

Both academic researchers and industry practitioners have applied MPC to optimize and control cooling systems. Google Research and Cloud applied MPC in a large-scale DC to control fan speeds and valve openings for each AHU, achieving approximately a 9 % reduction in server-floor cooling costs [22]. Study [23] deployed MPC in simulation for a multi-floor facility with five chillers and towers, improving each chiller's coefficient of performance and optimizing chilled-water storage to save 4–22 % of energy. Although the literature widely acknowledges MPC's benefits, its practical effectiveness hinges on prediction model accuracy. Study [17] identifies six key barriers to large-scale MPC deployment, foremost among them the need for user-friendly, control-oriented, accurate, and computationally efficient predictive models. Conventional MPC depends on precise mathematical models to forecast future system behaviors. However, accurately modeling complex or nonlinear systems poses significant challenges. In the meantime, developing physics-based prediction models for MPC is highly complex, requiring extensive expertise and considerable time investment. For example, Yang et al. developed a building dynamics model using a thermal resistance-capacitance framework integrated with MPC—a process that proved to be both highly complex and time-consuming [16]. They noted that physics-based building models necessitate detailed building information (e.g., structural dimensions, material properties, and system specifications) to construct models based on heat and mass transfer equations. This detailed information is crucial for accurately determining parameters such as thermal capacitances and resistances. Currently, the collection of such data and the subsequent model construction remain largely manual processes that demand significant expertise in building physics.

Recent advancements in machine learning (ML), particularly deep learning (DL), have spurred growing interest in employing neural networks (NNs) to construct system models. Neural networks effectively capture the complexities of systems by learning from historical data, enabling rapid and accurate predictions for real-time control. This data-driven approach approximates system dynamics without relying on explicit mathematical formulations. Moreover, operational data can be automatically acquired through building management systems (BMS), and training an ML-based model significantly reduces the human effort, time, and expertise required to develop an effective MPC system. Consequently, this alternative methodology not only enhances efficiency but also achieves energy savings comparable to traditional approaches [12].

Conventional machine learning approaches are fundamentally constrained by the scarcity of high-quality operational data, which limits their generalization capability and hinders real-world deployment. To overcome this barrier, researchers have begun embedding first-principles knowledge directly into neural-network architectures, thereby enhancing both predictive accuracy and data efficiency. This strategy has given rise to Physics-Informed Neural Networks (PINNs), which leverage neural networks as universal function approximators and incorporate governing equations— Partial Differential Equations (PDE) such as Navier–Stokes or Burgers equations—into the loss function [24–27]. Originally proposed as an alternative to grid-based solvers for forward and inverse PDE problems, PINNs improve predictive accuracy and generalization under scarce data, particularly in fluid mechanics where laws are naturally expressed as PDEs. The structure of PINN is illustrated in Fig. 2. A neural network with trainable weights and biases is trained on both measurement data and PDE residual points. Its loss function combines data and residual terms, and optimal parameters are obtained by minimizing this objective.

However, in building and cooling-system applications, the availability of relevant PDEs is more limited than in fluid-dynamics contexts. As a result, the direct application of traditional PINNs becomes challenging. To bridge this gap, Natale et al. introduced Physically Consistent Neural Networks (PCNNs) for building thermal modeling [28]. PCNNs consist of two components: a physical module that incorporates known heat-transfer and mass-balance relationships, and a black-box module in which neural networks capture nonlinear effects. This hybrid design achieved up to 40 % better performance than a classical physics-based resistance–capacitance model over three-day prediction horizons. Despite its innovative approach, PCNN struggles with accuracy and generalization in real-world thermal environments due to its reliance on empirically predetermined, static coefficients, which are costly to derive and limit adaptability to dynamic conditions. Building on this idea, we previous developed an Adaptive-PCNN (APCNN) that replacing traditional preset and fixed coefficients to reduce trial-and-error costs and increase flexibility [29]. Compared with PCNNs and conventional machine-learning models, APCNN demonstrates superior predictive accuracy. It is worth noting that APCNN is not a direct variant of PINNs but rather a physics-consistent hybrid framework. Both approaches integrate physical knowledge with deep learning, yet they differ fundamentally. APCNN is designed for systems lacking explicit

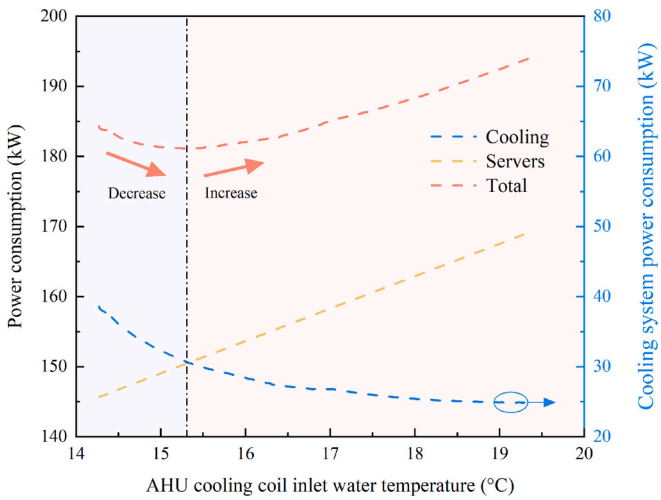

Fig. 3. Impact of AHU cooling-coil inlet water temperature on total energy consumption in an air-cooled EDC, based on 1-D simulations employing the equations from [15,33].

PDE formulations, leveraging algebraic thermodynamic and heat-transfer relations, whereas PINNs depend on governing PDEs. Moreover, APCNN enforces physical consistency through architectural constraints, while PINNs impose it via residual minimization.

Beyond developing PINNs and their variants for predictive modeling, researchers have integrated PINNs and their variants with MPC in diverse industrial applications. Hu et al. proposed physics-informed deep learning to predict temperature, humidity and CO_2 concentration, built a high-fidelity digital twin of a plant-factory microclimate and crop state. By integrating this twin with an MPC framework, they achieved a 13.41 % reduction in energy consumption and a 13.04 % decrease in resource use versus baseline [30]. Liang et al. employed PCNN models to predict indoor temperature dynamics in a multi-zone office building and coupled these predictions with MPC, yielding a 27 % reduction in peak demand and a 22 % decrease in energy use under a flat-rate tariff relative to baseline control [31]. Similarly, study [32] employed PINNs to develop chiller-plant energy models within an MPC framework, achieving a 23.2 % increase in energy efficiency over fixed-setpoint control in field experiments. However, their formulation focuses exclusively on minimizing chiller-plant energy without accounting for downstream impacts of supply-water temperature on IT equipment.

In DCs, servers—comprising both processors and their cooling fans—constitute the primary heat source. Processor power splits into dynamic and static (leakage) components, the latter of which grows exponentially with junction temperature; hence, at constant utilization, elevated chip temperatures drive up total energy use [34]. Moreover, elevated CPU temperatures also increase fan speeds and their associated power consumption. Haiqal [15] et al. implemented a Deep Reinforcement Learning controller for an air-cooled DC cooling system and showed that lowering the chiller supply water temperature yields net energy savings. A cooler chiller outlet lowers the server inlet-air temperature, which in turn reduces chip power draw and allows server fans to operate at lower speeds, further decreasing power consumption and total facility energy use. Leveraging these insights, we apply the formulas widely employed in the air-cooled DC literature [33] to evaluate how variations in the AHU cooling-coil inlet-water temperature affect overall system energy consumption. Specifically, we vary the chiller supply-water temperature while maintaining fixed AHU fan setpoints and CPU utilization and compute the power consumption of both the cooling system and the servers (CPUs plus server fans) as functions of inlet-water temperature (Fig. 3). Our results show that, at fixed CPU utilization, total power consumption exhibits a U-shaped trend: it decreases with rising water temperatures up to a certain point, then increases thereafter. As water temperature increases, cooling-system power consumption monotonically declines, whereas server power consumption steadily rises due to elevated CPU junction temperatures. Initially, the savings from reduced cooling demand outweigh the extra server energy, but beyond a critical water temperature, the increased server power dominates, causing a net rise in total energy use. These findings highlight the need for chiller-plant optimization strategies that explicitly balance heat-rejection efficiency with ITE thermal constraints, as neglecting AHU inlet-water temperature control can inadvertently increase overall system energy consumption.

1.2. Gaps and contributions

Our review identifies three critical gaps in the application of

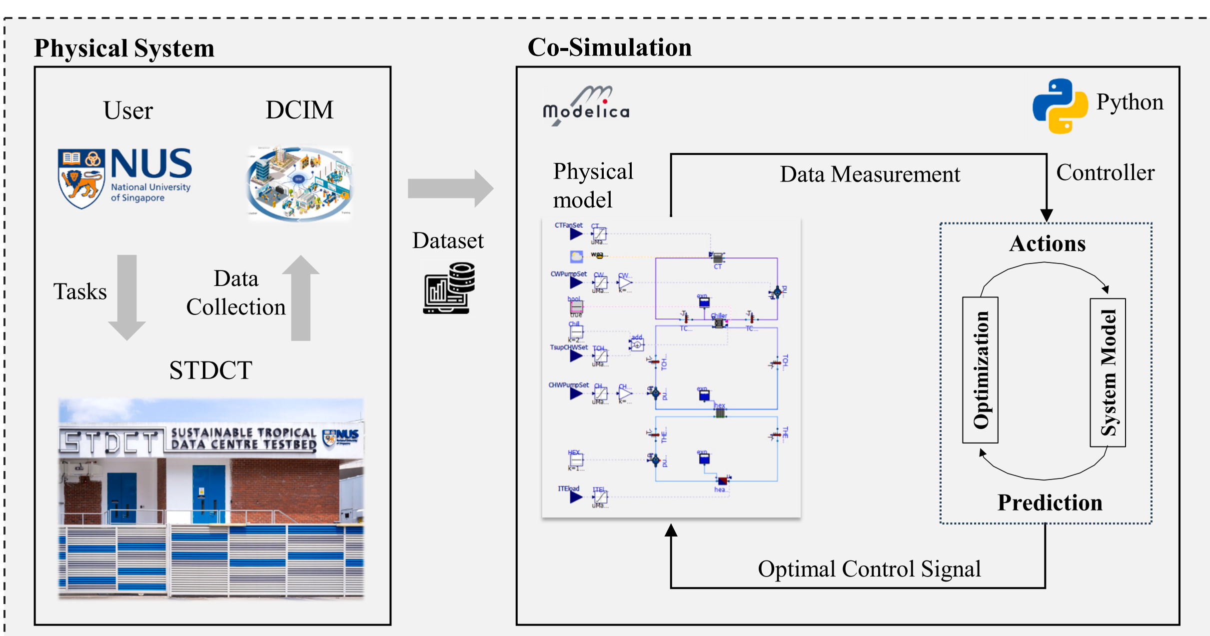

Fig. 4. Overview of the physical test facility and the co-simulation framework employed for DC cooling control using MPC approach.

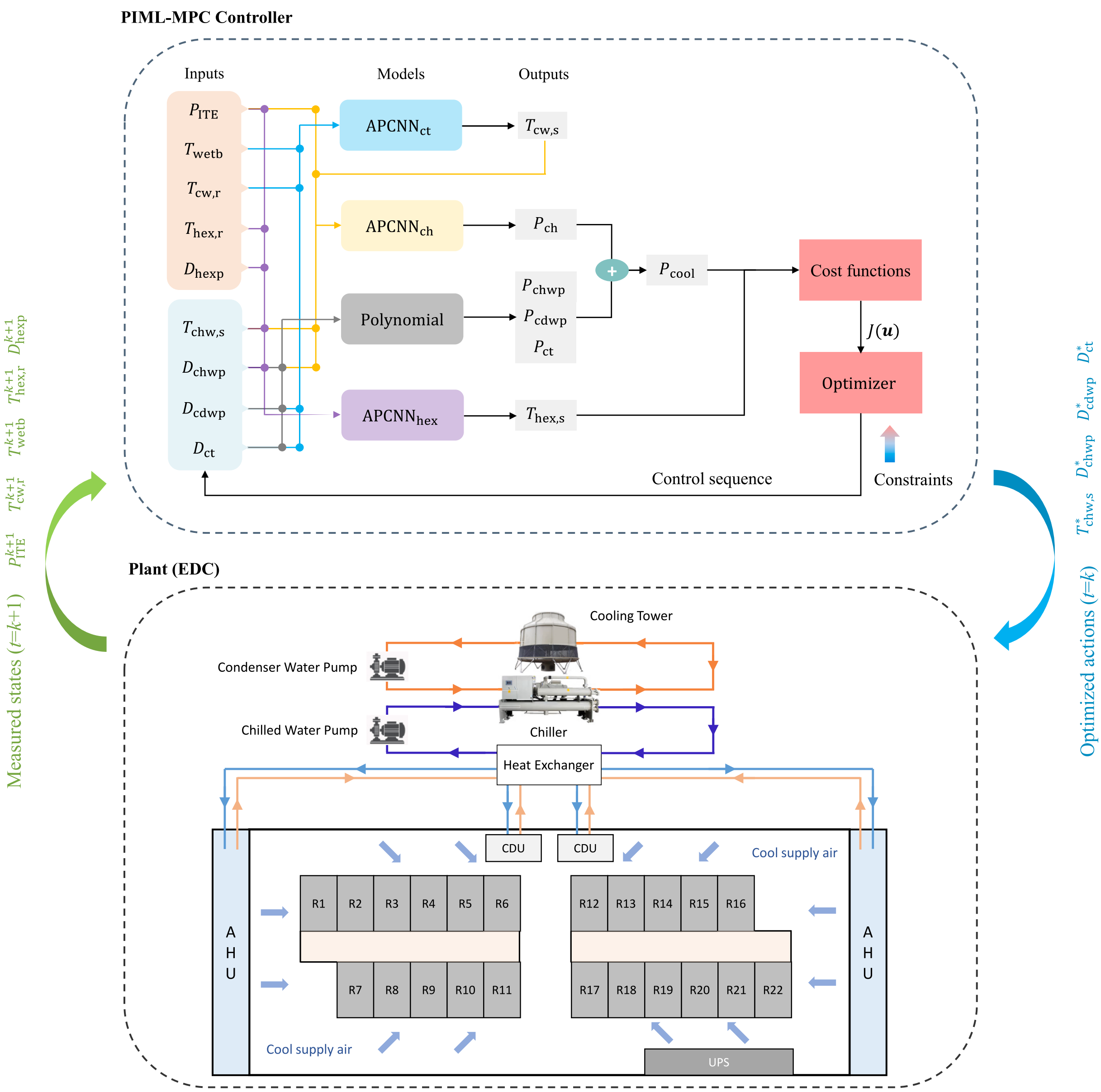

Fig. 5. Schematic representation of the PIML-MPC framework applied to the optimization of cooling system operations in EDCs. The Plant block illustrates the EDC data hall and its associated cooling infrastructure.

intelligent control to data centers. First, many MPC studies focus on commercial buildings or large-scale facilities, while compact EDCs—single-chiller, single-tower sites that enable low-latency, bandwidth-efficient, and privacy-preserving edge services—have been largely neglected. Second, although PINNs and their variants have enhanced MPC performance in other industries, they have not been systematically integrated into MPC frameworks for DCs/EDCs cooling that jointly enforce energy and temperature constraints. Third, despite demonstrated improvements in prediction accuracy and energy savings over fixed-setpoint methods, PINN-based controllers have seldom been benchmarked against MPC schemes employing conventional ML or time-series models, leaving their relative strengths and limitations insufficiently understood. Fourth, Because PINNs rely on ML techniques that are sensitive to random seed initialization, the extent to which seed-to-seed variability influences the performance and consistency of PINNs-based MPC remains unexplored.

Filling these gaps is essential for bringing MPC into real-world EDC operations. To conclude, this paper makes three main contributions:

1. We introduce a physics-informed machine learning (PIML) based-MPC framework for cooling system modeling and optimal control. Our PIML architecture employs Adaptive Physically Consistent Neural Networks (APCNN), which enforces exact physical laws as hard constraints.
2. Leveraging fundamental physical principles, we develop PIML predictors for the key cooling components of EDCs—namely cooling towers, chillers, and heat exchangers. These predictors are embedded within a MPC framework that enforces physical consistency over the entire control horizon while simultaneously accounting for both energy consumption and temperature constraints.
3. Using real EDC measurement data, we construct a high-fidelity simulation model. Simulation results demonstrate that PIML-MPC reduces energy consumption by 12.6 % compared to fixed setpoints control and achieves an additional 9.1 % savings under unseen high-load conditions, all while maintaining thermal regulation.
4. Benchmarking against three advanced ML and time-series MPC methods across ten random seeds demonstrates PIML-MPC's superior efficiency, thermal control, and minimal variability.

By addressing key gaps in existing research, these contributions advance the practical implementation of AI-driven MPC for DCs and EDCs management, delivering measurable gains in energy efficiency, system reliability, and scalability.

Fig. 4 illustrates the overall workflow of this study. Operational data from the Sustainable Tropical Data Centre Testbed (STDCT) were collected via the Data Center Infrastructure Management (DCIM) system. The raw dataset was then used to develop a simulation model in OpenModelica. This model was subsequently exported as an FMU file and integrated with Python. Within this environment, a PIML-MPC algorithm was implemented to dynamically adjust the cooling system settings during simulation, and its control performance was compared with that of the actual control method employed in STDCT. The remainder of this paper is organized as follows. Section 2 presents the proposed methodology. Section 2.1 outlines the overall PIML–MPC framework for EDC water-side control, while Section 2.2 introduces the APCNN prediction model, which serves as the core predictive component within the PIML-MPC architecture. Section 2.3 develops and calibrates component-level models for the heat exchanger, chiller, and cooling tower, embedding governing physical laws. Section 2.4 formalizes the MPC optimization problem, including the objective function, constraints, and solution strategy. Section 3 details the STDCT case study, including testbed configuration, system modeling, implementation procedures, and evaluation metrics. Section 4 presents and analyzes the results, and Section 5 concludes with a summary of our key findings and contributions.

2. Methodology

2.1. PIML-MPC framework

The EDC cooling system comprises two primary subsystems: an indoor portion—where data halls may employ AHUs, coolant distribution units (CDUs), liquid-cooled heat-sink modules, immersion cooling, or hybrid configurations—and a largely standardized outdoor plant. The outdoor plant typically includes a heat exchanger, chilled-water pump, chiller, condenser-water pump, and cooling tower. The heat exchanger transfers the ITE heat load from the indoor cooling loop to the outdoor plant. The chilled-water pump circulates the warmed water to the chiller, where a vapor-compression cycle extracts heat from the chilled-water circuit and rejects it to the condenser-water circuit. The condenser-water pump then conveys the heated water to the cooling tower, where evaporative cooling dissipates the rejected heat to the ambient air, lowering the condenser-water temperature before recirculation. This closed-loop arrangement ensures continuous and efficient extraction of ITE heat while maintaining the required temperature setpoints. In this work, we focus on the optimal control of the outdoor plant, with the primary objective of maintaining the supply water temperature entering the data hall within a prescribed range while minimizing the total power consumption of the cooling system. Therefore, the key variables can be classified into three categories: control inputs, states and disturbances.

1. Control inputs (manipulated variables):

- Chiller supply water temperature setpoint, $T^k_{\mathrm{chw,s}}$
- Chilled water pump setpoint, D^k_{chwp}
- Condensed water pump setpoint, D^k_{cdwp}
- Cooling tower fan speed, D^k_{ct}

2. Process states (predicted and regulated outputs):

- Power consumptions, P^k_{chwp}, P^k_{cdwp}, P^k_{ch}, P^k_{ct}
- Temperatures $T^k_{\mathrm{cw,s}}$, $T^k_{\mathrm{hex,s}}$

3. Disturbances (exogenous inputs affecting thermal behavior and energy use):

- ITE workload, P^k_{ITE}
- Outdoor air wet-bulb temperature, T^k_{wetb}

Building on this foundation, we propose a PIML-MPC framework specifically tailored to water-side control in EDCs, as shown in Fig. 5. The PIML component integrates APCNN models and classical polynomial regression, with each chosen according to the complexity of the input–output relationship. APCNNs are applied to parameters exhibiting complex, strongly nonlinear dependencies on multiple inputs, where the governing physical relationships are known. By embedding these physical laws into the network architecture, APCNNs combine the flexibility of neural networks with the ability to enforce physical consistency. In contrast, for simple one-to-one relationships—where each output is primarily determined by a single input—classical polynomial regression is adopted, as it delivers high fitting accuracy and strong generalization at low computational cost. Formally, at each control interval k, the MPC solves

$$\min_{u_1,\dots,u_N} \sum_{k=1}^{N} J^k, \tag{1}$$

$$\text{s.t.}\, \boldsymbol{u}^{min} \ll \boldsymbol{u}_k \ll \boldsymbol{u}^{max}, \tag{2}$$

$$T^{min}_{\mathrm{hex,s}} \ll T^k_{\mathrm{hex,s}} \ll T^{max}_{\mathrm{hex,s}}, \tag{3}$$

$$T^k_{\mathrm{chw,r}} - T^k_{\mathrm{chw,s}} \ll \Delta T^{max}_{\mathrm{chw,s}}, \tag{4}$$

$$T^k_{\mathrm{cdw,r}} - T^k_{\mathrm{cdw,s}} \ll \Delta T^{max}_{\mathrm{cdw,s}}, \tag{5}$$

where J is the combined cost term comprising power consumption and temperature violation. $\boldsymbol{u}_k = \left[T^k_{\mathrm{chw,s}}, D^k_{\mathrm{cdwp}}, D^k_{\mathrm{chwp}}, D^k_{\mathrm{ct}}\right]$ is the actions combination at the timestep k; $\boldsymbol{u}^{min}$ and $\boldsymbol{u}^{max}$ denote their respective bounds. $T^k_{\mathrm{chw,s}}$ and $T^k_{\mathrm{chw,r}}$ denote the respective chilled water supply and return temperatures, with $\Delta T^{max}_{\mathrm{chw,s}}$ specifying the maximum allowable temperature difference. $T^k_{\mathrm{cdw,s}}$ and $T^k_{\mathrm{cdw,r}}$ denote the respective condensed water supply and return temperatures, with $\Delta T^{max}_{\mathrm{cdw,s}}$ defining the maximum allowable difference. The chilled-water temperature difference is constrained because an excessively large value deteriorates heat-transfer performance and disrupts stable load matching, thereby reducing system reliability and energy efficiency. Similarly, the condenser-water temperature difference must also be limited, as an excessive value impedes heat rejection, disturbs water distribution, and increases condenser pressure, ultimately compromising system safety and efficiency. $T^{min}_{\mathrm{hex,s}}$ and $T^{max}_{\mathrm{hex,s}}$ define the permissible supply-water temperature range to avoid overcooling and ensure safe server operation. The preceding introduction demonstrates that focusing solely on heat-load transfer—without constraining the supply water temperature—can increase downstream equipment energy use. In the absence of this constraint, a controller could simply raise the chiller setpoint and lower fan and pump setpoints—thereby reducing immediate cooling power at the expense of downstream heat-removal capacity. We next introduce the APCNN model and detail the development of component models for the heat exchanger, chiller, and cooling tower.

2.2. Adaptive physically consistent neural networks

Operational datasets from real-world plants are often limited, constraining the generalizability of purely data-driven prediction models. Embedding established physical laws within the learning process can mitigate this limitation by enhancing model robustness under conditions

Table 1
Activation functions for bounded adaptive coefficients.

Name	Equation	Range
Sigmoid	$\sigma(x) = \frac{1}{1+e^{-x}}$	$(0,1)$
Tanh	$\tanh(x) = \frac{1-2e^{-2x}}{1+2e^{-2x}}$	$(-1,1)$
ReLU	$\mathrm{ReLU}(x) = \max(0,x)$	$(0,\infty)$

of data scarcity. Accordingly, we propose an APCNN approach for EDC modeling, in which both the network architecture and the constraint mechanisms are customized to reflect the physical characteristics and data availability of each target variable [29]. Consider a target variable Y that governed by three inputs x_1, x_2, and x_3. The APCNN prediction $\widehat{Y}$ is expressed as the sum of the physical-module output E and black-box module outputs B:

$$\widehat{Y} = E + B. \tag{6}$$

The physical module encodes domain knowledge via analytical expressions or structural relationships derived from physical laws, with adaptive coefficients constrained to respect properties such as sign consistency, monotonicity, or boundedness. The black-box module captures complex, residual dynamics that are nonphysical or analytically intractable. For the physical module, one prescribes a specific functional form consistent with known physics. For instance, if the predicted variable is known to depend linearly on two of the inputs, one may set

$$E = a \bullet x_1 + b \bullet x_2, \tag{7}$$

where adaptive coefficients a and b represent are produced by the network backbone but constrained—via appropriate activation functions (Table 1)—to enforce $\frac{\partial Y}{\partial x_1} > 0$ (e.g. ReLU on a) or any other required property. Both the coefficients and the black-box output are generated jointly by a shared feature extractor f_w:

$$a, b, B = f_w(x_1, x_2, x_3). \tag{8}$$

This shared backbone increases the model's capacity to capture interactions among inputs while ensuring that the physical component adheres strictly to known relationships. The APCNN is trained by minimizing the mean squared error (MSE):

$$\mathscr{L} = \frac{1}{M}\sum_{i=1}^{M}\left(Y^i - \widehat{Y}^i\right)^2. \tag{9}$$

By hard-wiring the physics module, this architecture enforces exact compliance with known laws, while the black-box module captures unmodeled effects such as sensor noise or secondary couplings. Although Eq. (7) illustrates a simple linear combination, more sophisticated physically motivated forms—such as polynomials, products, or exponentials—can readily be substituted. The effectiveness of APCNN critically depends on the appropriate integration of physical knowledge, and specific designs for individual EDC components are presented in the following section. A detailed mathematical proof is provided in our previous work [29]. Notably, the activation functions used to enforce coefficient constraints are applied in the final layer of the shared backbone f_w. The APCNN provides robust, physics-informed predictions that strike a balance between flexibility and physical fidelity across diverse modeling tasks.

2.3. Component models

2.3.1. Heat exchanger temperature model

A heat exchanger (HEX) is a device that transfers heat between two fluids (liquid or gas) without allowing them to mix. In the EDC cooling system, the HEX transfers heat from the indoor cooling loop—which absorbs heat from the servers—to the outdoor plant loop. In this study, we adopt the Effectiveness–Number of Transfer Units (ε-NTU) method to derive mathematical expressions that capture the physical relationships between the output and input parameters. Using this widely accepted approach, we obtain the following relationships:

$$Q_{\max} = C_{\min}\left(T_{\text{hex,r}} - T_{\text{chw,s}}\right), \tag{10}$$

$$Q_{\text{real}} = \varepsilon \bullet Q_{\max} = \varepsilon C_{\min}\left(T_{\text{hex,r}} - T_{\text{chw,s}}\right), \tag{11}$$

$$T_{\text{hex,s}} = T_{\text{hex,r}} - \frac{Q_{\text{real}}}{C_{\text{hex}}} = T_{\text{hex,r}} - \varepsilon\frac{C_{\min}}{C_{\text{hex}}}\left(T_{\text{hex,r}} - T_{\text{chw,s}}\right), \tag{12}$$

$$T_{\text{hex,s}} = \varepsilon\frac{C_{\min}}{C_{\text{hex}}}T_{\text{chw,s}} + \left(1 - \varepsilon\frac{C_{\min}}{C_{\text{hex}}}\right)T_{\text{hex,r}}, \tag{13}$$

where $Q_{\max}$ and Q_{real} denote the maximum possible and the actual heat transfer rates, respectively; $C_{\max}$ and $C_{\min}$ represent the maximum and minimum heat capacity rates of the heat exchanger for the two fluids; ε is the heat exchanger efficiency, defined as the ratio of the actual heat transfer to the maximum possible heat transfer. In an ideal situation, that is, with infinite heat transfer area, ε can approach 1, but the heat transfer area is limited. Consequently, we have $0 < \varepsilon < 1$ and $0 < \frac{C_{\min}}{C_{\text{hex}}} \ll 1$, which implies:

$$0 < \varepsilon\frac{C_{\min}}{C_{\text{hex}}} < 1. \tag{14}$$

Based on the energy balance equation, we get:

$$P_{\text{ITE}} = \dot{m}_{\text{hex}}c_p\left(T_{\text{hex,r}} - T_{\text{hex,s}}\right), \tag{15}$$

Thus, the return water temperature from data hall to the heat exchanger can be expressed as:

$$T_{\text{hex,r}} = T_{\text{hex,s}} + \frac{P_{\text{ITE}}}{\dot{m}_{\text{hex}}c_p}, \tag{16}$$

By integrating the above equations, we obtain:

$$T_{\text{hex,s}} = T_{\text{chw,s}} + \left(\frac{1}{\varepsilon\frac{C_{\min}}{C_{\text{hex}}}} - 1\right)\frac{P_{\text{ITE}}}{\dot{m}_{\text{hex}}c_p}, \tag{17}$$

Based on these physical laws and observations from actual processes, we propose the following APCNN model:

$$T^k_{\text{hex,s}} = E^k + B^k, \tag{18}$$

$$E^k = T^k_{\text{chw,s}} + \left(\frac{1}{a^k} - 1\right)\frac{P_{\text{ITE}}}{\dot{m}_{\text{hex}}c_p}. \tag{19}$$

where E^k and B^k denote the outputs of the physical and black-box modules, respectively; a^k represents the adaptive coefficient. According to the above equations, $a^k = \varepsilon\frac{C_{\min}}{C_{\text{hex}}}$ is constrained to lie between 0 and 1. Moreover, in practical engineering applications, it is often observed that the real heat transfer process exhibits systematic deviations (e.g., due to temperature sensor errors, heat dissipation losses, or bypass flow effects) from the ideal behavior. To accommodate these deviations, we introduce the offset B^k, which is constrained to remain small over the operating range. To enforce these constraints, we use the Sigmoid activation function to limit a^k within (0,1), where γ is set to 1.0. In addition, we adopt the Tanh activation function to restrict B^k within $(-\beta, +\beta)$, where β is set to 1.0. This yields the following equations. Both bounds are enforced via activation functions:

$$a^k = \gamma \bullet \sigma(out_1), B^k = \beta \bullet \tanh(out_2), \tag{20}$$

where

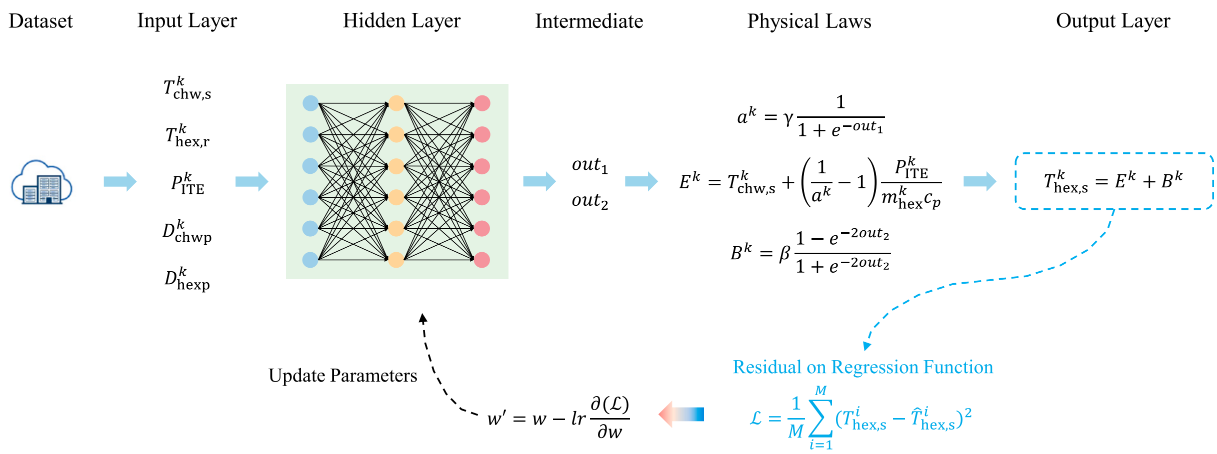

Fig. 6. Schematic of the $\text{APCNN}_{\text{hex}}$ model.

$$out_1, out_2 = f_w\left(T_{\text{chw,s}}, T_{\text{hex,r}}, P_{\text{ITE}}, D_{\text{chwp}}, D_{\text{hexp}}\right), \tag{21}$$

where out_1 and out_2 are the two outputs of the base fully connected network f_w. The loss function for the $\text{APCNN}_{\text{hex}}$ model is defined as:

$$\mathcal{L}_{\text{hex}} = \frac{1}{M}\sum_{i=1}^{M}\left(T^i_{\text{hex,s}} - \hat{T}^i_{\text{hex,s}}\right)^2, \tag{22}$$

where $\hat{T}^i_{\text{hex,s}}$ represents the predicted water temperature delivered from the heat exchanger to the data hall. Collectively, these mathematical expressions encapsulate the fundamental thermodynamic principles governing heat exchanger performance. By embedding these physical formulas directly into the neural network architecture, the model is structurally regularized to ensure that the sensitivity of the predicted heat exchanger outlet temperature with respect to the input variables conforms to the expected physical behavior. Finally, we got a built APCNN model for HEX supply water temperature modeling (see Fig. 6):

$$T_{\text{hex,s}} = \text{APCNN}_{\text{hex}}\left(T_{\text{chw,s}}, T_{\text{hex,r}}, P_{\text{ITE}}, D_{\text{chwp}}, D_{\text{hexp}}\right). \tag{23}$$

2.3.2. Chiller power model

A chiller is a machine used to generate low-temperature water through the refrigeration process, which in turn cools the data center's air conditioning system or directly dissipates heat from the cabinet. Its core components—the evaporator, compressor, condenser, and expansion valve—work together to control the refrigerant cycle and optimize heat exchange. In the refrigeration cycle, heat Q_L is extracted from the low-temperature (refrigeration) space at temperature T_L and rejected to the high-temperature (heat rejection) space at temperature T_H. Under idealized conditions, the chiller cycle approximates a reversible Carnot cycle consisting of two isothermal and two isentropic processes. This ideal cycle attains the maximum theoretical coefficient of performance (COP), defined as

$$\text{COP} = \frac{Q_L}{W} = \frac{1}{\frac{T_H}{T_L} - 1}, \tag{24}$$

where W is the net work input required for the cycle. In data center cooling, the ITE load represents the heat that must be removed. Thus, the chiller's power consumption is

$$P_{\text{ch}} = \frac{P_{\text{ITE}}}{\text{COP}}. \tag{25}$$

In reality, COP varies with operating conditions: higher evaporation (chilled-water) temperatures, lower condensation (cooling-water) temperatures, and operation near design load all tend to increase COP. To capture these dependencies while preserving physical interpretability, we scale a nominal design-condition COP by an adaptive coefficient a^k, and employ a data-driven module to learn residual corrections. Specifically, at k timestep we decompose the predicted chiller power as

$$P^k_{\text{ch}} = E^k + B^k, \tag{26}$$

where E^k is the physics-informed term and B^k is the black-box adjustment. We define

$$E^k = \frac{P_{\text{ITE}}}{a^k \bullet \text{COP}_{\text{ref}}}, \tag{27}$$

where COP_{ref} is the nominal COP at design conditions. To ensure a^k remains in the plausible range $(0, \gamma)$, we apply a Sigmoid activation:

$$a^k = \gamma \bullet \sigma(out_1), \tag{28}$$

Simultaneously, we bound B^k within $(-\beta, +\beta)$, via a Tanh activation:

$$B^k = \beta \bullet tanh(out_2), \tag{29}$$

The two pre-activations out_1 and out_2 are produced by a shared fully connected network f_w:

$$out_1, out_2 = f_w\left(T_{\text{chw,s}}, T_{\text{cw,s}}, P_{\text{ITE}}, D_{\text{chwp}}\right). \tag{30}$$

The loss function for the APCNN_{ch} model is defined as:

$$\mathcal{L}_{\text{ch}} = \frac{1}{M}\sum_{i=1}^{M}\left(P^i_{\text{ch}} - \hat{P}^i_{\text{ch}}\right)^2. \tag{31}$$

Together, these components define our final APCNN model:

$$P_{\text{ch}} = \text{APCNN}_{\text{ch}}\left(T_{\text{chw,s}}, T_{\text{cw,s}}, P_{\text{ITE}}, D_{\text{chwp}}\right). \tag{32}$$

This formulation combines first-principles thermodynamics with data-driven corrections, delivering both interpretability and flexibility in modeling chiller power under varying operating conditions.

2.3.3. Cooling tower temperature model

The cooling tower lowers the temperature of circulating water by coupling sensible- and latent-heat exchange with ambient air. Warm water is distributed over the fill media and descends as a thin film (or rivulets), while ambient air is drawn in from below (or the sides) and flows counter-currently (or in crossflow) to the water. As the two streams interact, a fraction of the water evaporates—absorbing its latent heat of vaporization—and the remaining liquid is cooled. The air's wet-bulb temperature T_{wetb}, sets the theoretical lower bound for evaporative cooling: water can only evaporate when its temperature exceeds T_{wetb}, and the CT outlet water temperature $T_{\text{cw,s}}$, asymptotically approaches but never falls below T_{wetb}. The approach temperature T_{approach} is defined as:

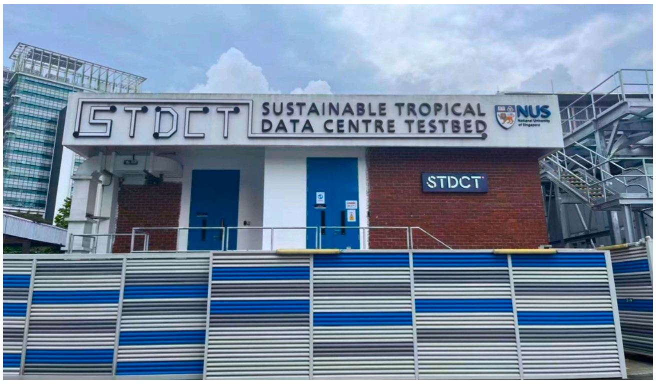

Fig. 7. Sustainable tropical data centre testbed.

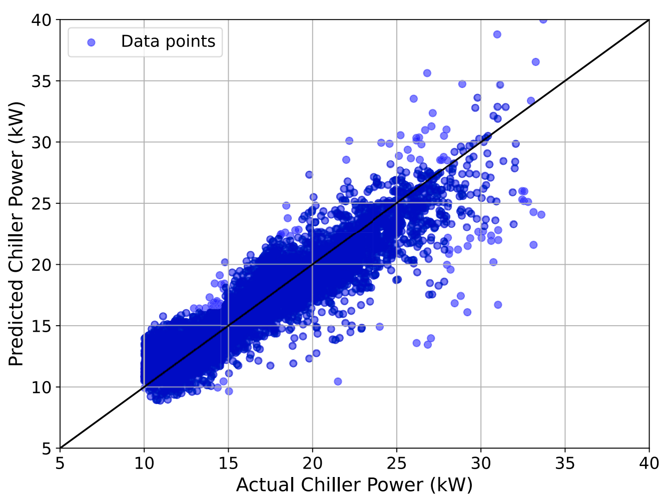

Fig. 8. Regression analysis of chiller power consumption based on 14,613 data points.

Table 2
The results of the simulated model compared to the measured data from STDCT.

Metrics	Simulation data	Measured data	Difference/%
P_{ct}/kW	0.85	0.82	3.66
P_{cdwp}/kW	5.46	5.30	3.02
P_{ch}/kW	12.88	13.63	5.50
P_{chwp}/kW	2.79	2.89	3.46

$$T_{\text{approach}} = T_{\text{cw,s}} - T_{\text{wetb}}, \tag{33}$$

which quantifies the proximity of the CT outlet temperature to the wet-bulb limit. The temperature drops across the tower, i.e., the difference between the hot-water inlet and cooled-water outlet, is defined as the range temperature:

$$T_{\text{range}} = T_{\text{cw,r}} - T_{\text{cw,s}}, \tag{34}$$

where $T_{\text{cw,r}}$ is cooling tower inlet water temperature. Combining eqs. (31) and (32) yields:

$$T_{\text{approach}} = T_{\text{cw,r}} - T_{\text{wetb}} - T_{\text{range}}, \tag{35}$$

with the physical constraint:

$$T_{\text{approach}} \in \left(0, T_{\text{cw,r}} - T_{\text{wetb}}\right), \tag{36}$$

In practice, T_{approach} is often obtained by fitting a high-order polynomial to site-specific experimental data. However, such purely empirical fits require extensive operating-condition data, which may be unavailable. Instead, recognizing that a cooling tower is essentially a specialized heat exchanger between air and water, we employ the classical Number-of-Transfer-Units (NTU) method for a first-principles estimate of the approach temperature:

$$T_{\text{approach}} = \left(T_{\text{cw,r}} - T_{\text{wetb}}\right) \bullet e^{-NTU}, \tag{37}$$

where $NTU = \frac{UA}{C_{min}}$ is the dimensionless heat-transfer capacity. This form guarantees T_{approach} remains within the physically admissible bounds in Eq. (36). Because NTU varies with operating conditions (water and air flow rates, fill geometry, fouling, etc.), we introduce a data-driven correction on top of the physics-based estimate. Specifically, we model the outlet temperature at time step k as:

$$T^{k}_{\text{cw,s}} = E^{k} + B^{k}, \tag{38}$$

Table 3
Performance metrics for two control methods under seen conditions.

Metrics	Baseline	PIML-MPC	Absolute Improvement	Relative Improvement
P_{cool}/kWh	15,352.1	13,422.0	1930.1	12.60 %
P_{chwp}/ kWh	1504.2	575.2	929.0	61.80 %
P_{ch}/ kWh	12,745.5	12,339.7	405.8	3.20 %
P_{cdwp}/ kWh	491.2	207.2	284.0	57.80 %
P_{ct}/ kWh	611.1	299.9	311.2	50.90 %
$\max(T_{\text{hex,s}})$/ °C	17.1	18.7	–	–
$\overline{T}_{\text{hex,s}}$/ °C	16.8	18.5	–	–
TVR/%	0.0	0.0	–	–
CSE	0.137	0.119	0.018	13.10 %

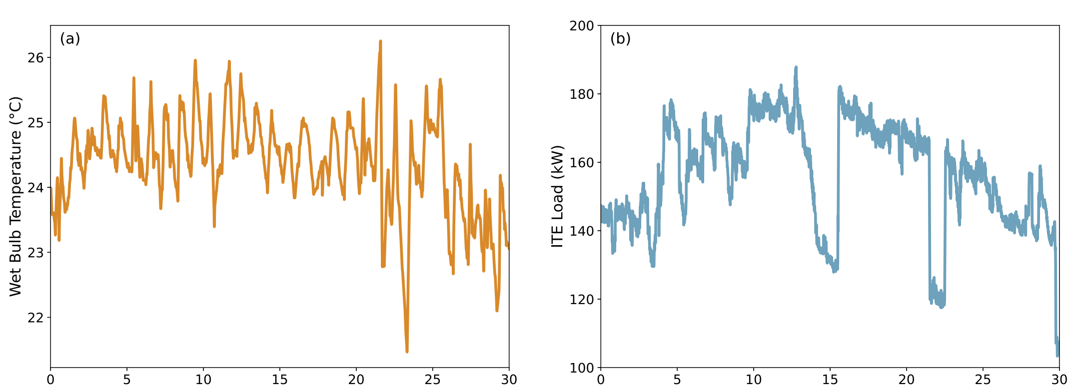

Fig. 9. External disturbances: (a) fluctuations of outdoor air wet-bulb temperatures and (b) variations in ITE load data.

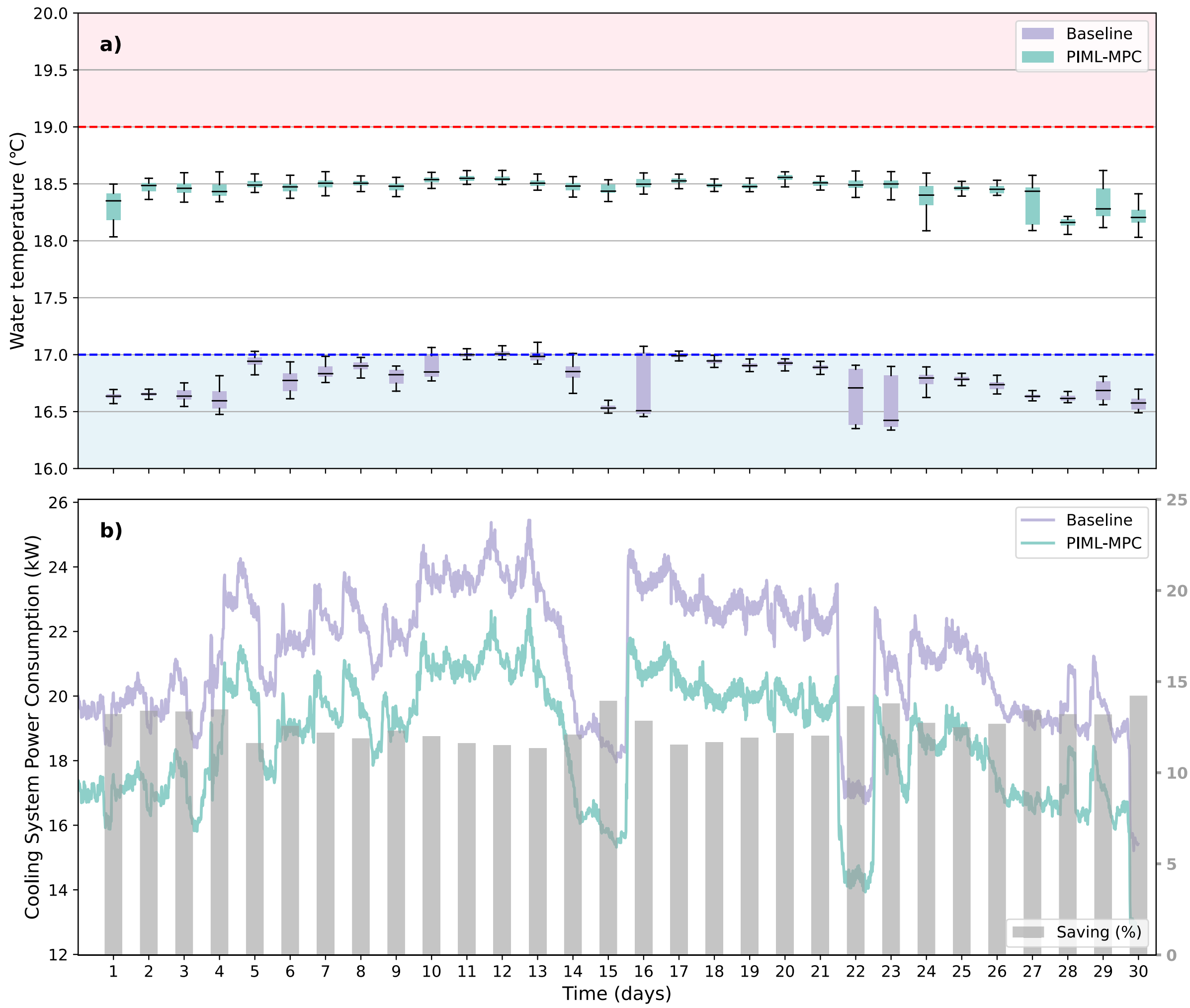

Fig. 10. Comparative performance of baseline and PIML-MPC (seed 40) under familiar conditions: (a) temperature regulation; (b) cooling power consumption and relative daily energy savings.

where E^k is the physics-informed component and B^k is a data-driven correction. We define E^k using an adaptive exponent a^k that modulates the effective NTU:

$$E^k = T^k_{\text{wetb}} + \left(T^k_{\text{cw,r}} - T^k_{\text{wetb}}\right) \bullet e^{-a^k}. \tag{39}$$

To ensure remains a^k within $(0, \gamma)$ and B^k within $(-\beta, +\beta)$, we apply

$$a^k = \gamma \bullet \sigma(out_1), B^k = \beta \bullet tanh(out_2). \tag{40}$$

where

$$out_1, out_2 = f_w\left(T_{\text{cw,r}}, T_{\text{wetb}}, D_{\text{ct}}, D_{\text{cdwp}}\right). \tag{41}$$

The model is trained by minimizing the MSE over a dataset of size M:

$$\mathscr{L}_{\text{ct}} = \frac{1}{M}\sum_{i=1}^{M}\left(T^i_{\text{cw,s}} - \widehat{T}^i_{\text{cw,s}}\right)^2. \tag{42}$$

Finally, we got a built APCNN model for CT supply water temperature modeling:

$$T_{\text{ct,s}} = \text{APCNN}_{\text{ct}}\left(T_{\text{cw,r}}, T_{\text{wetb}}, D_{\text{ct}}, D_{\text{cdwp}}\right). \tag{43}$$

This hybrid architecture ensures that the bulk of the prediction follows first-principles heat transfer behavior (via the NTU-based term), while the neural correction captures unmodeled nonlinearities, fouling effects, and external disturbances.

2.4. Optimization problem formulation

At each time step k, the MPC objective combines two cost terms: power consumption and temperature violation. The overall cost J^k can be expressed as:

$$J^k = \lambda_1 \bullet J^k_{\text{power}} + \lambda_2 \bullet J^k_{\text{temp}}. \tag{44}$$

The power consumption cost J^k_{power} aggregates the electrical power drawn by all four outdoor-plant components at time k:

$$J^k_{\text{power}} = \left(P^k_{\text{chwp}} + P^k_{\text{ch}} + P^k_{\text{cdwp}} + P^k_{\text{ct}}\right). \tag{45}$$

The temperature violation cost is defined as follows:

$$J^k_{\text{temp}} = \left(T^k_{\text{hex,s}} - T_{\text{sp}}\right)^2, \tag{46}$$

where $T^k_{\text{hex,s}}$ represent the water temperature from heat exchanger supply to the data hall at k timestep; T_{sp} denotes the setpoint water temperature. The weighting factors λ_1 and λ_2 balance the MPC controller's multiple objectives. In this work, they are both chosen as 1.

The MPC operates in a closed-loop configuration, where the optimization problem is re-solved at each control interval using real-time measurements of system states and exogenous variables. The control interval is set to 15 min, with a control horizon of one interval and a prediction horizon of two intervals (30 min). At each step, the optimizer

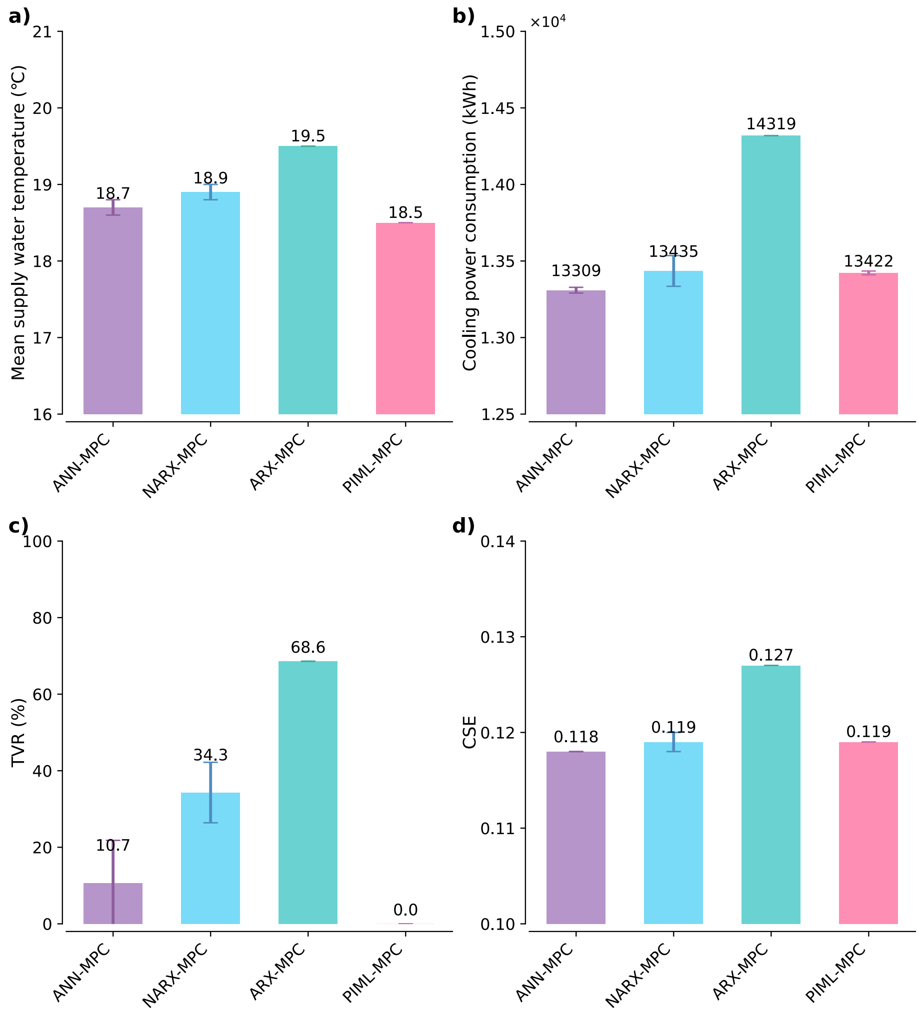

Fig. 11. Comparative performance of four control strategies under familiar operating conditions. Values represent the average values derived from 10 trained models.

computes a two-step control sequence; however, only the first control input is applied before the system state is updated and the optimization is executed again in the next interval. Because exogenous variables are not forecasted, the computed sequence is optimized solely with respect to the current operating conditions. This formulation ensures feasibility under rapidly varying ITE loads and weather conditions, while also substantially reducing the computational burden, since the optimization cost scales almost linearly with the length of the prediction horizon.

Since the cooling system dynamics are nonlinear, a nonlinear optimizer is required to solve the underlying MPC problem. Purely global optimization methods can effectively identify global optima but are generally too computationally expensive for real-time applications, while purely gradient-based methods converge faster but risk becoming trapped in local optima if initial guesses are not well chosen. To overcome these limitations, we follow the hybrid optimization strategy proposed in [35]. In the first stage, a coarse grid search is conducted around the previous control actions to generate a finite set of candidate solutions for the four decision variables. Each candidate is evaluated using the three component models (heat exchanger, cooling tower, and chiller), and its corresponding cost function is computed. The three candidates with the lowest cost are then selected as initial solutions for the second stage, as this number provides a balance between exploration diversity and computational efficiency. In this stage, local refinement is carried out using the Sequential Least Squares Quadratic Programming (SLSQP) algorithm, a gradient-based constrained nonlinear optimization method that leverages the differentiability of the surrogate models to refine the coarse grid solutions while strictly enforcing operational and physical constraints. This global-local framework combines the robustness of grid-based global exploration with the efficiency of local optimization, thereby enabling high-quality solutions with manageable computational overhead. The hybrid optimization procedure was implemented in Python using the SLSQP algorithm from SciPy's optimization module.

3. Case study

3.1. STDCT testbed

The Sustainable Tropical Data Centre Testbed, located at the National University of Singapore (NUS), is a data center testbed in the tropics, dedicated to advancing sustainable and efficient solutions for high-temperature, high-humidity environments in Singapore and beyond (Fig. 7). A schematic diagram of the STDCT data hall and cooling system is shown in Fig. 5. The data hall houses 22 rack-mounted Dell servers operated by NUS IT, which ensures that the IT load accurately reflects actual operational demand. STDCT employs a hybrid cooling system that integrates both air and liquid cooling technologies to enhance thermal management efficiency. The system comprises AHUs, CDUs, a water-cooled chiller, a cooling tower, and associated pumps and pipes. The AHUs are positioned along both sides of the data hall to efficiently dissipate heat generated by ITE and ancillary systems. Two CDUs are deployed within the hall to regulate, control, and distribute

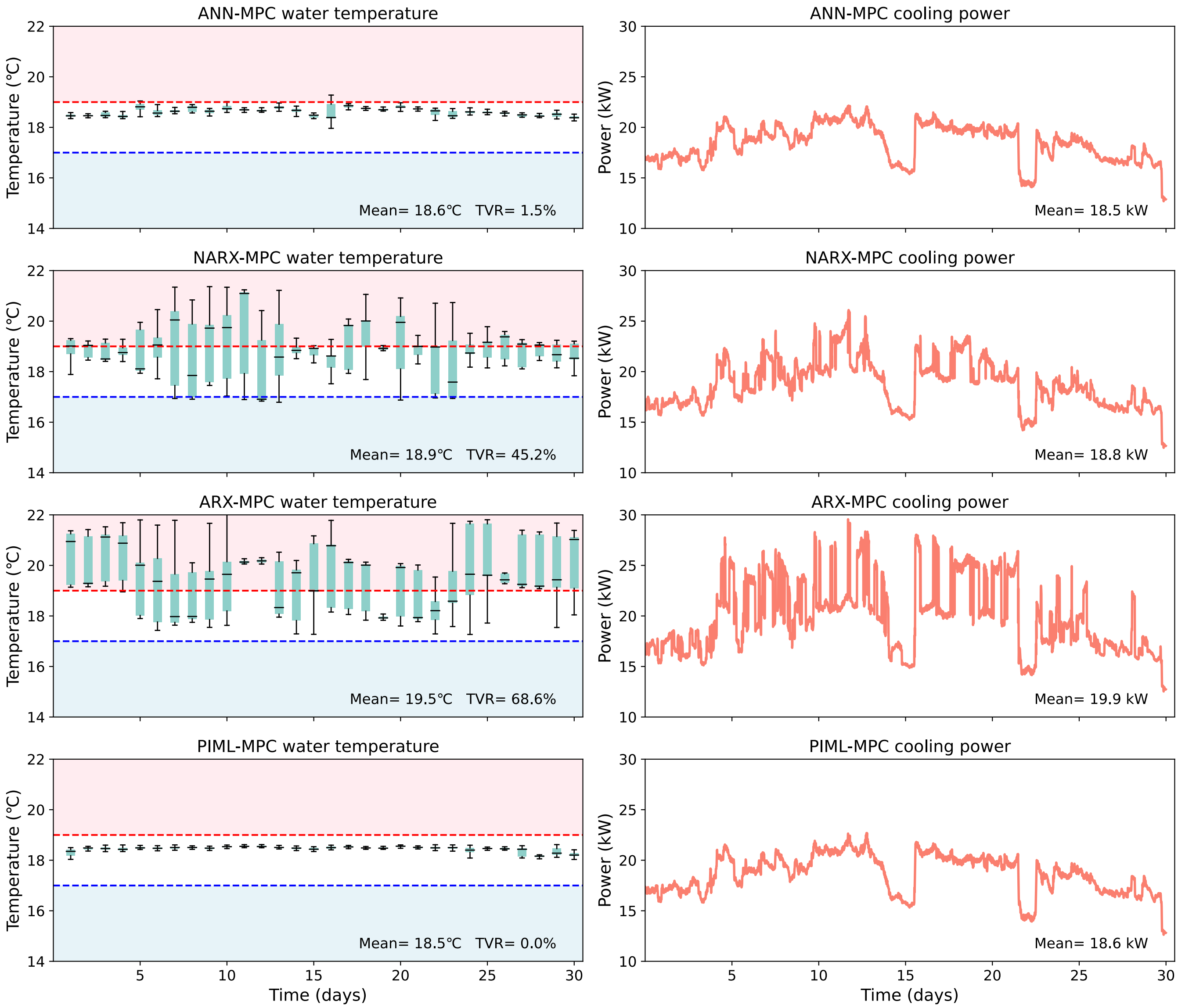

Fig. 12. Performance comparison of four control methods under familiar conditions (seed 40).

coolants, ensuring optimal cooling system performance. These CDUs facilitate effective heat transfer from the CPUs and GPUs to the chilled water system. To further enhance thermal efficiency, Hot Aisle Containment is implemented. In this configuration, the hot air outlet side of the server cabinets is fully enclosed to form a dedicated heat channel, preventing the dispersion of hot air into the cooler areas of the data center. The confined hot return air is directed through an independent, sealed upper system to the AHUs, where it is cooled by the cooling coils and then redistributed to the data hall via the AHU fans. In addition, the system incorporates a water-cooled centrifugal chiller manufactured by Trane, which operates using R134a refrigerant. This chiller delivers a nominal cooling capacity of 571 kW and achieves a nominal coefficient of performance (COP) of 6.42. Complementing the chiller, a counterflow cooling tower provided by Truwater is utilized; it is designed to handle a heat load of 756,000 kcal/h, reducing the water temperature from an inlet of 40 °C to an outlet of 34 °C under an ambient wet bulb temperature of 30.2 °C. The cooling tower fan's rated capacity is approximately 7.2 kW. The system maintains a total water flow rate of 35 L/s, thereby ensuring consistent and efficient heat exchange throughout the cooling cycle. In the STDCT, fixed-setting control is employed, whereby the system operates with predefined, constant parameters rather than dynamically adjusting based on real-time feedback. Settings derived from prior operational experience, ensure stable system performance without requiring continuous adjustments.

3.2. STDCT cooling system modeling

OpenModelica is an open-source, Modelica-based modeling and simulation environment widely used in both industrial and academic applications. It provides extensive library support, including a variety of models for data center cooling systems [36]. In this work, we employ OpenModelica to model the outdoor cooling system illustrated in Fig. 5. The model is designed to capture the essential thermal and fluid-dynamic behaviors of a typical EDC outdoor cooling loop. The system architecture is composed of two primary subsystems: the Condenser Water Loop and the Chilled Water Loop. The Condenser Water Loop includes a cooling tower and a condenser water pump. In the cooling tower, heat is dissipated through water evaporation, which reduces the temperature of the condensate prior to its recirculation. The condenser water pump then ensures continuous flow by returning the cooled condensate to the system for reuse. The Chilled Water Loop comprises a water-cooled chiller and a chilled water pump. Within the chiller, a refrigeration cycle extracts heat from the water, transferring it to the cooling water. This heat is subsequently removed by the cooling tower.

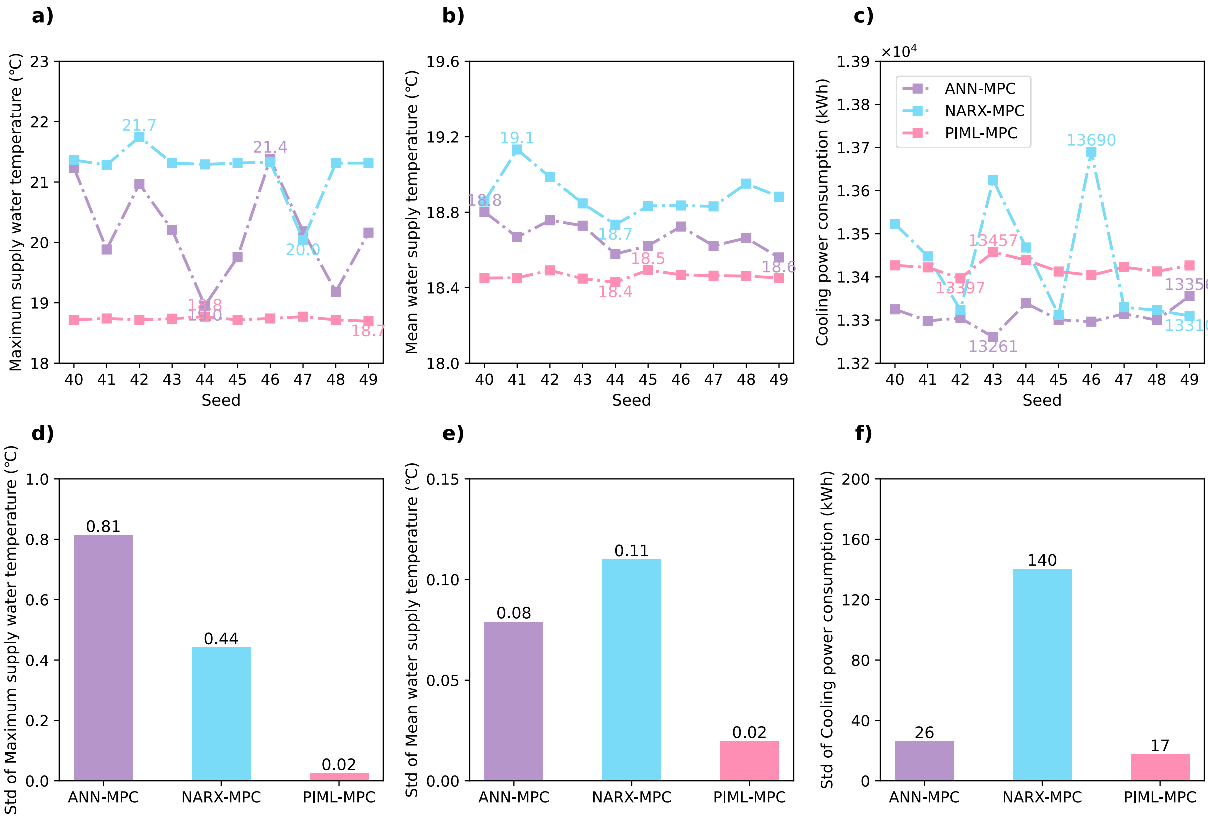

Fig. 13. Performance of the three control strategies under seen conditions, aggregated over ten independent trials (random seeds) in panels (a)–(c), with their corresponding standard deviations in panels (d)–(f).

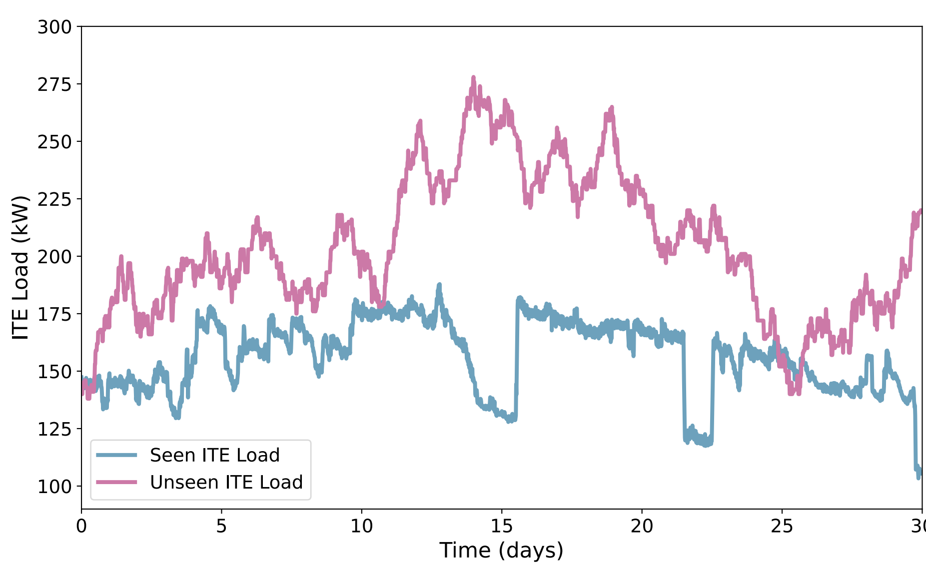

Fig. 14. Temporal profile of ITE load disturbances, where the blue trace corresponds to the training dataset and the red trace corresponds to the unseen dataset used to evaluate model generalization. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The chilled water pump circulates the cooled water throughout the system, facilitating efficient heat exchange and maintaining the desired temperature for air conditioning or industrial processes.

Specifically, the simulation model integrates four key components: the chilled water pump, the condenser water pump, the water-cooled chiller, and the cooling tower. Two variable-speed pumps are utilized within the system. The power consumption of the pumps is calculated as the product of the water flow rate and the pressure drop, divided by the pump's total efficiency. Additionally, the chiller is modeled as a water-cooled electric gray box—a common approach in data center simulations—using the condenser water entering temperature as a primary parameter. This model captures both the chiller's thermal performance and the compressor's power consumption by incorporating three performance curves to address off-reference operating conditions:

Cooling Capacity Function of Temperature Curve (CapFT):

$$CapFT = a_1 + a_2 T_{cw,ls} + a_3 T_{cw,ls}^2 + a_4 T_{cond,e} + a_5 T_{cond,e}^2 + a_6 T_{cw,ls} T_{cond,e}, \tag{47}$$

Table 4
Performance metrics for two control methods under unseen conditions.

Metrics	Baseline	PIML-MPC	Absolute Improvement	Relative Improvement
P_{cool}/kWh	19,863	18,047.2	1815.8	9.10 %
P_{chwp}/ kWh	1504.2	1017.0	487.2	32.40 %
P_{ch}/ kWh	17,256.5	16,445.6	810.9	4.70 %
P_{cdwp}/ kWh	491.2	226.9	264.3	53.80 %
P_{ct}/ kWh	611.1	357.8	253.3	41.40 %
$\max(T_{hex,s})$/ °C	18.2	19.0	–	–
$\overline{T}_{hex,s}$/ °C	17.3	18.6	–	–
TVR/%	0.0	1.0	–	–
CSE	0.135	0.122	0.013	9.60 %

where $T_{cw,ls}$ is the leaving chilled water setpoint temperature; $T_{cond,e}$ is the entering condenser fluid temperature. For a water-cooled condenser this will be the water temperature returning from the condenser loop (e.g., leaving the cooling tower).

Energy Input to Cooling Output Ratio Function of Temperature Curve (EIRFT):

$$EIRFT = b_1 + b_2 T_{cw,l} + b_3 T_{cw,l}^2 + b_4 T_{cond,e} + b_5 T_{cond,e}^2 + b_6 T_{cw,l} T_{cond,e}, \tag{48}$$

where $T_{cw,l}$ is the leaving chilled water temperature.

Energy Input to Cooling Output Ratio Function of Part Load Ratio Curve (EIRFPLR):

$$EIRFPLR = c_1 + c_2 PLR + c_3 PLR^2, \tag{49}$$

where PLR, or Part Load Ratio, is calculated by dividing the cooling load by the available cooling capacity of the chiller. Finally, the power consumption of the chiller, denoted as $P_{chiller}$, is expressed as follows:

$$P_{ch} = \frac{P_{ch,ref}}{COP_{ref}} \bullet CapFT \bullet EIRFT \bullet EIRFPLR, \tag{50}$$

where $P_{ch,ref}$ denotes the nominal capacity of chiller; COP denotes coefficient of performance. The cooling tower operates at variable speeds, and its model is formulated using existing empirical data based on the York correlations, which is widely used [33,37]. These correlations are integrated into the standard cooling tower model within OpenModelica.

Measured STDCT data were employed to calibrate and validate the model parameters. Fig. 8 depicts the regression analysis of chiller power consumption using 14,613 observations, yielding a mean absolute error (MAE) of 0.79 and an R-squared value of 0.87. For further validation, a representative operating condition was selected—namely, a heat load of 103.0 kW, cooling tower fan speed at 50 %, condenser water pump operating at 80.0 %, cooling water pump at 77 %, and chiller outlet water temperature of 17.2 °C. Table 2 presents a comparison between the simulated model results and the measured data from STDCT. The findings indicate that the cooling system model effectively replicates the real data center's power consumption, demonstrating both the accuracy and reliability of the simulation model.

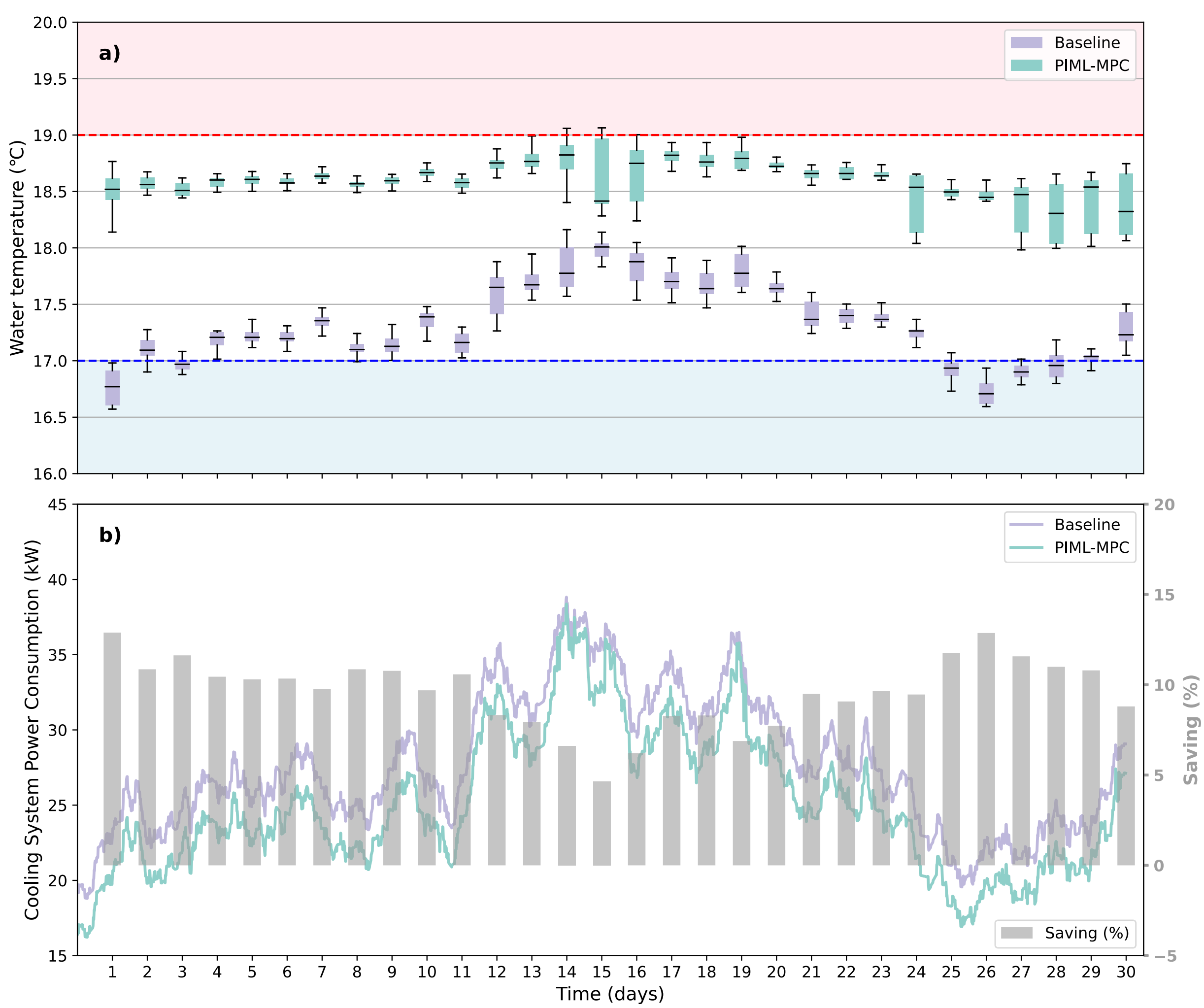

Fig. 15. Performance comparison of baseline control and PIML-MPC (seed 40) under unseen conditions.

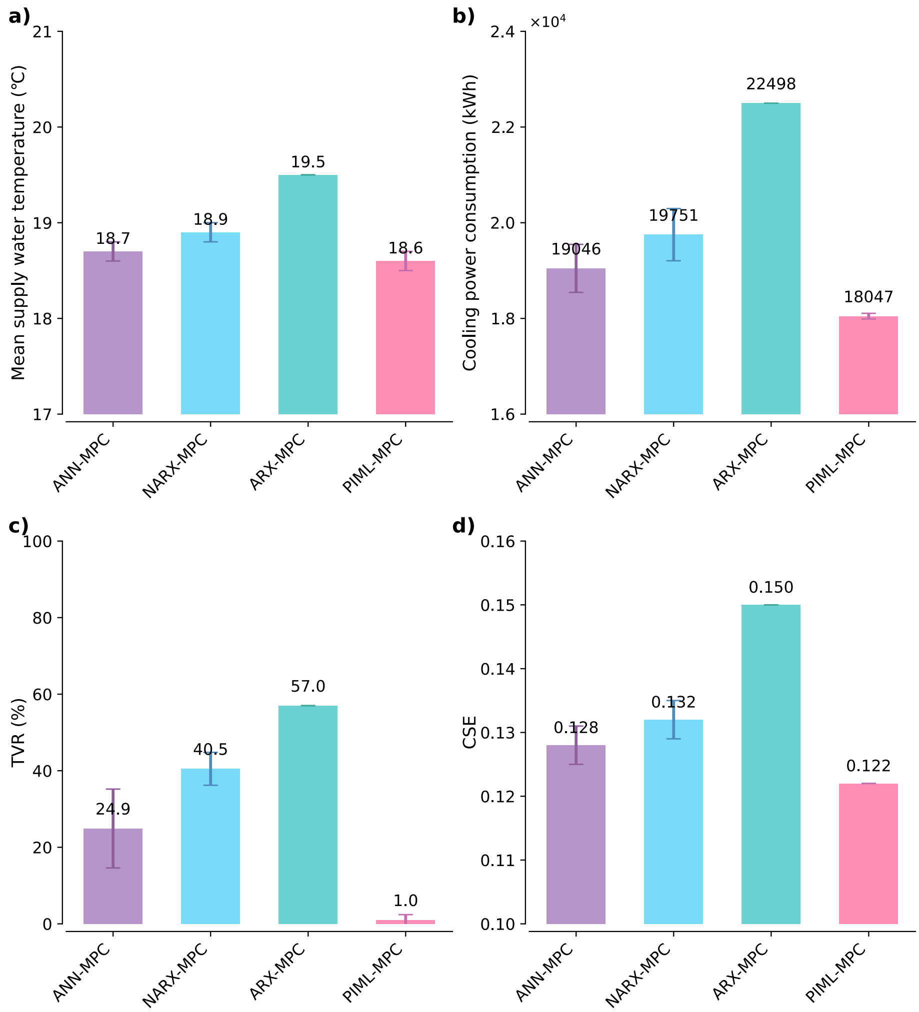

Fig. 16. Comparative performance of four control strategies under familiar operating conditions. Values represent the average values derived from 10 trained models.

3.3. Implement details

Following ASHRAE's recommended data-hall temperature range of 18 °C to 27 °C and based on Dell company's analysis, the optimal server inlet temperature lies between 24 °C and 26 °C [38], where total DC power consumption is minimized. Accordingly, we constrain the supply-water temperature to 18 ± 1 °C. Within this window, we systematically evaluate and compare the energy-saving performance of each control strategy. Thus, we set T_{sp}=18 °C in Eq. (46), and set $T^{min}_{\text{hex,s}} = 17$ °C and $T^{max}_{\text{hex,s}} = 19$ °C. The maximum allowable temperature differences, $\Delta T^{max}_{\text{chw,s}}$ and $\Delta T^{max}_{\text{cdw,s}}$, are set to 8 °C to prevent excessively large temperature differentials, which could impair heat transfer and compromise system stability. In the STDCT cooling system, each of the four control variables operates within predefined limits. The corresponding operating ranges and step sizes used in the MPC optimization are summarized in Table C.1. The PIML models were employed as predictive components within the MPC framework, which require representative training data. To this end, 30 days of historical data were collected from the data center model operating under a fixed-settings control scheme with a 15-min sampling interval, yielding 2880 samples. However, because only a single set of control parameters was used during data acquisition, the action space was not fully explored. To overcome this limitation, we applied the exploration method described in [22], augmenting the dataset with an additional 480 samples. The final dataset therefore comprised 3360 samples, all recorded at 15-min intervals.

To further validate our PIML-MPC framework, we benchmark it against three widely adopted time-series and machine-learning MPC methods. First, several studies have integrated an Autoregressive model with exogenous inputs (ARX) into an MPC controller (ARX-MPC) for DC cooling systems [22,39]. Second, a Nonlinear Autoregressive model with exogenous inputs (NARX) coupled with an artificial neural network—forming a NARX-MPC scheme—has demonstrated improved handling of system nonlinearities in building cooling applications [35]. Third, a purely data-driven ANN-MPC approach has been proposed, wherein an artificial neural network is embedded directly within the MPC loop without any physics constraints [40]. Additional details on the benchmark models are provided in Appendix A and B. For a fair and rigorous comparison, all predictive models were trained on the same dataset using identical initialization; likewise, the cost function, constraints, and optimization algorithm were standardized across all MPC implementations.

3.4. Evaluation metrics

To quantify the performance of each control algorithm in terms of thermal regulation and energy consumption, we employ two complementary metrics: Temperature Violation Rate (TVR) and Cooling System Efficiency (CSE). Let $\left\{T^k_{\text{hex,s}}\right\}_{k=1}^{N}$ denote the sequence of supply water temperatures sampled at N discrete time instants. We define the set of violation indices:

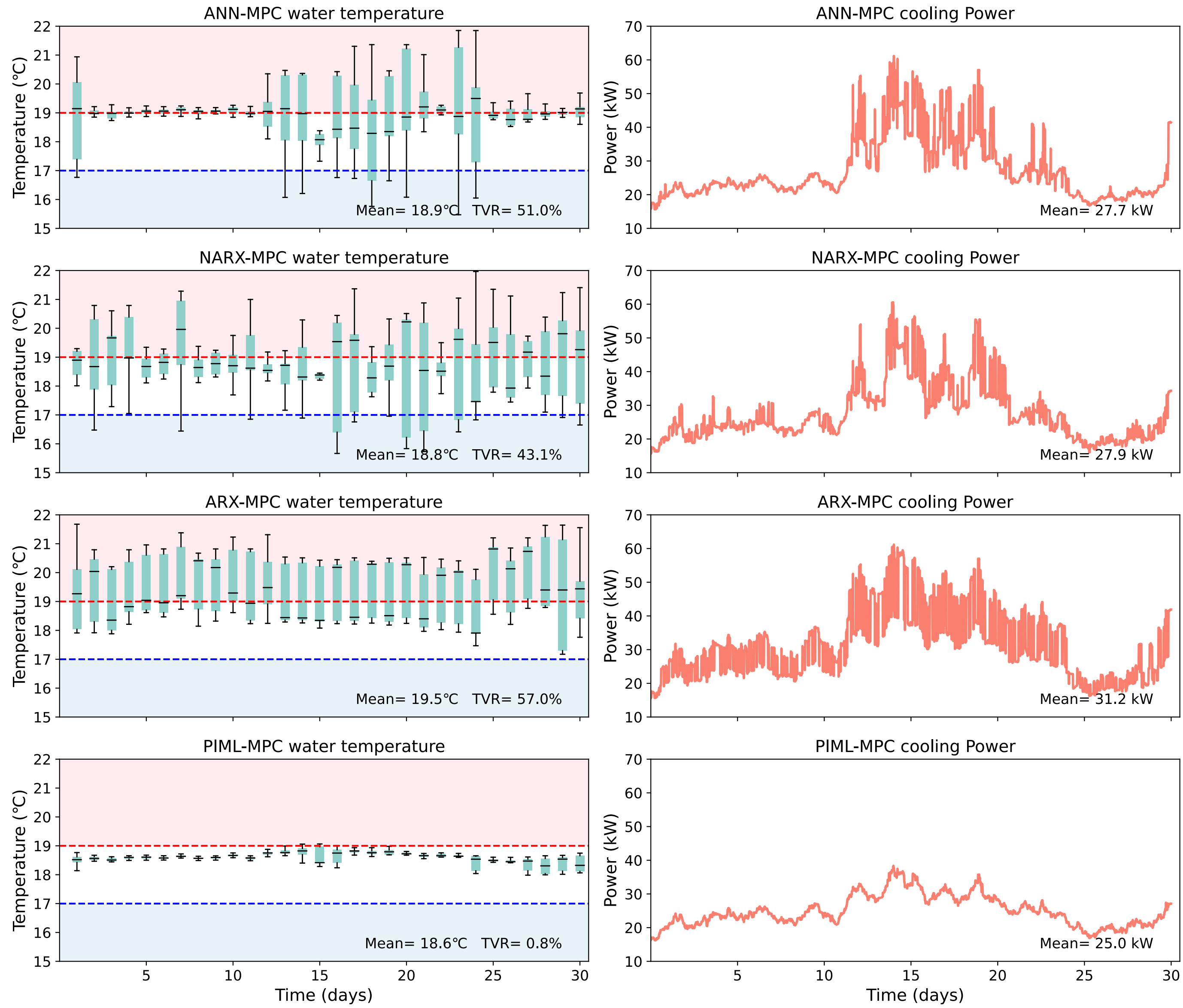

Fig. 17. Performance comparison of four control methods under unseen conditions (seed 40).

$$\mathscr{F} = \left\{k \in \{1, \cdots, N\} \,|\, T^{k}_{\mathrm{hex,s}} > T_{th}\right\}. \tag{51}$$

The TVR is then given by

$$\mathrm{TVR} = \frac{|\mathscr{F}|}{N} = \frac{1}{N}\sum_{k=1}^{N} 1\left(T^{k}_{\mathrm{hex,s}} > T_{th}\right), \tag{52}$$

where $1(\cdot)$ is the indicator function. A lower TVR indicates stricter compliance with the upper temperature threshold and more effective thermal management. In our experiments, the nominal setpoint is 18 °C with a ± 1 °C tolerance. Thus, the temperature threshold $T_{th} = 19$ °C. Any reading above 19 °C constitutes a violation, while temperatures below 17 °C are not penalized, as they pose no risk of equipment damage.

The CSE is defined as the ratio of the cooling system's power consumption P_{cool} to the IT equipment's power consumption P_{ITE}:

$$\mathrm{CSE} = \frac{P_{\mathrm{cool}}}{P_{\mathrm{ITE}}}. \tag{53}$$

A lower CSE reflects a more energy-efficient cooling architecture. By jointly considering TVR and CSE, we obtain a balanced view of each controller's ability to maintain safe operating temperatures while minimizing energy use. This dual-metric framework guides the design and selection of control strategies that enhance both sustainability and cost-effectiveness in DC cooling.

For each ML based MPC controller, we trained ten independent instances using distinct random seeds for both model initialization and the split between training and validation sets. Reported performance metrics represent the average over these ten cases. Specifically, for any given metric m, we compute the sample mean $\overline{m}$ and standard deviation as:

$$\overline{m} = \frac{1}{N}\sum_{i=1}^{N} m^{i}, std = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(m^{i} - \overline{m}\right)^{2}}. \tag{54}$$

To quantify uncertainty, we then construct a two-sided 95 % confidence interval:

$$\overline{m} \pm 2.262 \bullet \frac{std}{\sqrt{N}}. \tag{55}$$

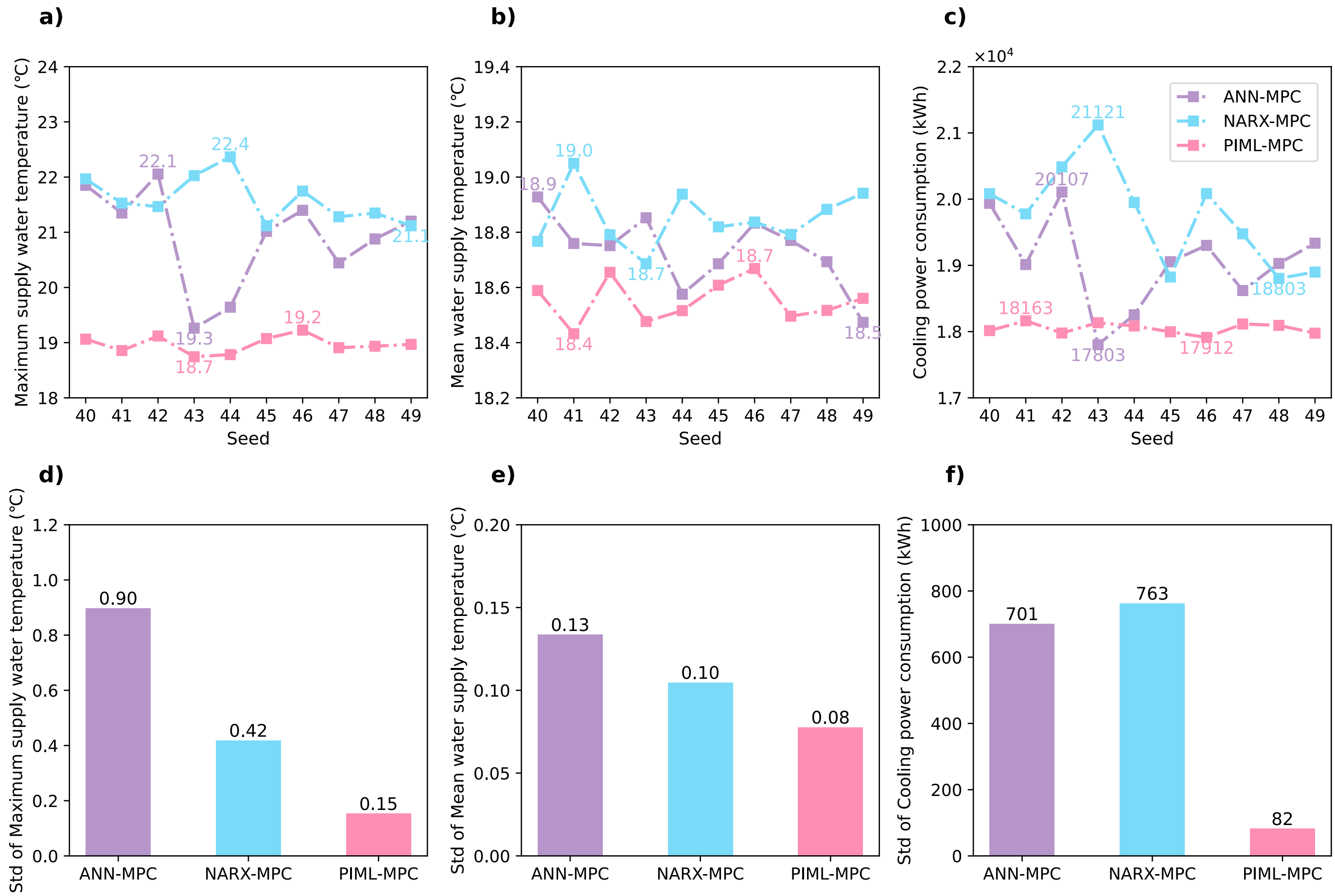

Fig. 18. Performance of the three control strategies under unseen conditions, aggregated over ten independent trials (random seeds) in panels (a)–(c), with their corresponding standard deviations in panels (d)–(f).

4. Results and discussion

Initially, we compare the performance of our proposed PIML-MPC approach against four control methods, thoroughly evaluating its enhancements in temperature regulation and energy efficiency under familiar conditions. Recognizing that real-world DC environments are often suboptimal and that operational data for training predictive models is typically limited, we acknowledge the generalization challenges inherent in many machine learning methods, which frequently struggle with novel scenarios. Thus, we also assess the robustness of the PIML-MPC method under previously unseen conditions. Under two scenarios, we first compare PIML-MPC method with baseline (fixed setting control) which is used in the real EDC. In the baseline method, four controllable parameters are maintained at predetermined constant values derived from prior operational experience, ensuring stable system performance even under high ITE load demand. In addition, we assess its performance relative to three alternative schemes: ANN-MPC, ARX-MPC, and NARX-MPC.

4.1. Controller performance in seen scenarios

In this section, we assess the proposed PIML-MPC strategy under familiar scenarios. Fig. 9 depicts the external disturbances that significantly impact both the thermal environment and energy consumption but remain beyond direct control. Specifically, Fig. 9 (a) presents the outdoor ambient temperature, while Fig. 9 (b) shows the ITE load data collected at STDCT. Each experiment spanned a 30-day period.

4.1.1. Comparison with baseline

Table 3 summarizes the performance comparison between the baseline and the PIML-MPC strategies under seen conditions. Under PIML-MPC, total cooling power consumption decreases from 15,352.1 kWh to 13,422.0 kWh—an absolute saving of 1930.1 kWh, corresponding to a 12.60 % relative reduction. This improvement is driven primarily by auxiliary equipment: the chilled-water pump energy use falls from 1504.2 kWh to 575.2 kWh (saving 929.0 kWh, 61.80 %), and the condenser-water pump consumption decreases from 491.2 kWh to 207.2 kWh (284.0 kWh saved, 57.80 %). The cooling tower fan power also drops from 611.1 kWh to 299.9 kWh (311.2 kWh saved, 50.90 %), reflecting more precise control of fan speed and water flow. Although the chiller remains the largest energy consumer, its consumption still declines from 12,745.5 kWh to 12,339.7 kWh—an absolute saving of 405.8 kWh (3.20 %). Finally, the CSE improves from 0.137 under the baseline to 0.119—a gain of 0.018, or a 13.10 % relative enhancement—underscoring the comprehensive efficiency benefits of the PIML-MPC framework. Thermally, the PIML-MPC controller raises the maximum HEX supply-water temperature from 17.1 °C to 18.7 °C and the mean supply temperature from 16.8 °C to 18.5 °C, while maintaining a TVR of 0.0 %, thereby fully upholding all safety constraints. It worth noting that work [32] proposed a physics-informed neural network for optimal chiller-plant control—incorporating both structural and trend-based prior knowledge—and demonstrated a 23.2 % improvement in energy efficiency by resetting the control setpoint. However, their approach focuses exclusively on heat load removal and neglects the impact of supply-water temperature, which can increase downstream equipment energy consumption.

Fig. 10 illustrates the heat exchanger supply water temperature, total cooling system power consumption, and the daily energy-saving ratio over the same 30-day interval. Fig. 10 (a) presents the temperature regulation performance of the two controllers. The dashed horizontal lines at 19 °C and 17 °C indicate the upper temperature limit (beyond which overheating may occur) and the lower limit (below which overcooling is undesirable), respectively. The baseline strategy maintains water temperatures below 17 °C approximately 91.8 % of the time, effectively ensuring cooling but resulting in higher energy consumption. In contrast, the PIML-MPC strategy consistently operates at temperatures above 17 °C and closer to the 19 °C upper threshold without exceeding it, thereby achieving more efficient temperature control. Notably, both approaches maintain a TVR of 0.0 %, indicating reliable adherence to thermal constraints. Fig. 10 (b) compares the total cooling system power consumption with the daily energy savings of the two control methods. The PIML-MPC approach consistently delivers superior energy efficiency, achieving daily energy savings ranging from 11.3 % to 14.2 %. Furthermore, on days with low ITE loads, the energy savings with the PIML-MPC strategy become even more pronounced. This occurs because the baseline strategy employs constant settings that are optimized for high-load conditions, leading to overcooling and unnecessary energy expenditure under low-load conditions.

4.1.2. Comparison with existing MPC controller

Fig. 11 compares the performance of four MPC variants under previously seen operating conditions. Although the supply water temperature was constrained between 17.0 °C and 19.0 °C, only PIML-MPC successfully maintained the temperature within this range, achieving a mean supply-water temperature of 18.5 °C with a TVR of 0 %. In contrast, ARX-MPC exhibited the poorest thermal regulation, with the supply-water temperature rising to 19.5 °C and a TVR of 68.6 %, indicating frequent violations of the upper temperature limit. ANN-MPC and NARX-MPC also failed to maintain strict compliance, recording mean supply-water temperatures of 18.7 °C and 18.9 °C, respectively, along with relatively high TVRs of 10.7 % and 34.3 %—suggesting frequent boundary violations despite near-optimal mean values. These findings demonstrate that the physics-informed prediction model underlying PIML-MPC enables the most accurate and constraint-compliant temperature control among all compared strategies. In terms of energy consumption, ANN-MPC achieved the lowest cooling energy use at 13309.4 kWh, followed closely by PIML-MPC at 13422.0 kWh, NARX-MPC at 13435.1 kWh, and ARX-MPC at 14318.9 kWh. Consequently, ANN-MPC attained a CSE of 0.118, while PIML-MPC and NARX-MPC both reached 0.119, and ARX-MPC recorded 0.127. Although ANN-MPC consumes slightly less energy, it fails to maintain temperature constraints as effectively as PIML-MPC, exhibiting noticeable violations. In contrast, PIML-MPC achieves both strict thermal compliance and high energy efficiency. Overall, these findings demonstrate that PIML-MPC delivers the most balanced performance—minimizing energy use while fully eliminating temperature violations, thereby ensuring robust, constraint-compliant operation and superior overall efficiency compared with both machine-learning-based and time-series-based MPC approaches.

Beyond the aggregate analysis, we examined daily control performance using a fixed random seed. Fig. 12 shows the HEX supply water temperature and cooling power consumption over 30 days (seed 40) for four control strategies. Under identical operating conditions, only PIML-MPC maintained the supply water temperature within the specified bounds; the other controllers exhibited substantial oscillations in both temperature and power consumption. PIML-MPC effectively modulated all four control inputs, consistently keeping temperatures within limits and minimizing thermal and power fluctuations. Among the remaining methods, ANN-MPC achieved the best trade-off between energy savings and temperature regulation, showing the lowest energy consumption and the second-lowest TVR. In contrast, ARX-MPC and NARX-MPC performed poorly in both aspects, exhibiting severe daily fluctuations and overall control failure.

Fig. 13 illustrates the performance of three MPC strategies under seen conditions, evaluated over ten independent trials (random seeds 40–49). Panels (a)–(c) present the temperature-regulation metrics and total cooling energy consumption for each controller, while panels (d)–(f) display the corresponding standard deviations. Notably, PIML-MPC confined its maximum supply water temperature to a narrow 18.7–18.8 °C range (*std* = 0.02 °C), whereas ANN-MPC and NARX-MPC exhibited much larger variability, with *std* = 0.81 °C and *std* = 0.44 °C, respectively. Similarly, PIML-MPC maintained its mean supply water temperature at approximately 18.4 °C (*std* = 0.02 °C), in contrast to standard deviations of 0.08 °C for ANN-MPC and 0.11 °C for NARX-MPC. In terms of total cooling energy, PIML-MPC achieved a minimum std. of 17 kWh, compared with fluctuations of 26 kWh for ANN-MPC and 140 kWh for NARX-MPC.

Overall, the results indicate that PIML-MPC exhibits the lowest run-to-run variability among ML-based controllers. This reduced sensitivity to random initialization stems from its embedded physical constraints, which narrow the model's hypothesis space and limit parameter variation. By guiding training toward physically consistent solutions, these constraints reduce divergence across different initial seeds. Consequently, the model's predictions—and, by extension, the MPC control actions—remain more stable than those of purely data-driven approaches such as ANN-MPC and NARX-MPC. In practical deployments, this consistency eliminates the need for extensive re-training or hyperparameter tuning, simplifies system commissioning, accelerates deployment, and reduces operational risk by ensuring predictable behavior across diverse real-world environments. The ARX model is omitted from this comparison, as its closed-form least-squares solution produces fully deterministic results, independent of random seed initialization.

4.2. Controller robustness in unseen scenarios

The performance of ML-based models is heavily influenced by the quantity and quality of the training data. In typical DC operations, however, the full range of potential variations in parameters—such as ITE load—is often not captured, resulting in prediction models being trained on inherently limited datasets. To comprehensively assess our PIML-MPC framework under truly novel conditions, we evaluated its performance using ITE power levels that extended beyond those observed during training. Fig. 14 illustrates these broader, unobserved operating conditions, which were monitored over a 30-day period to test the model's adaptability across a wider spectrum of disturbances.

4.2.1. Comparison with baseline

Table 4 presents the comparison between the baseline control and the PIML-MPC strategy over the same 30-day period. Under the PIML-MPC scheme, total cooling energy consumption falls from 19,863.0 kWh to 18,047.2 kWh, an absolute saving of 1815.8 kWh (9.10 %). This reduction is driven primarily by auxiliary equipment: chilled-water pump power demand decreases from 1504.2 kWh to 1017.0 kWh (487.2 kWh saved, 32.40 %), condenser-water pump use is cut from 491.2 kWh to 226.9 kWh (264.3 kWh saved, 53.80 %), and cooling-tower fan energy drops from 611.1 kWh to 357.8 kWh (253.3 kWh saved, 41.40 %). Under the baseline controller, the chiller consumed 17,256.5 kWh; by contrast, PIML-MPC reduced this to 16,445.6 kWh—an absolute saving of 810.9 kWh (4.70 %)—demonstrating that even the primary cooling load benefits substantially from advanced predictive control. Finally, the CSE index improves from 0.135 under the baseline to 0.122, a relative gain of 9.60 %, showing that the PIML-MPC framework optimizes overall system efficiency by shifting load away from fixed-setpoint operation toward dynamic, prediction-driven scheduling. Thermally, the PIML-MPC controller raises the maximum HEX supply-water temperature from 18.2 °C to 19.0 °C and the mean supply-water temperature from 17.3 °C to 18.6 °C, with the TVR slightly increasing from 0.0 % to 1.0 %, demonstrating that safety constraints remain effectively satisfied. Across ten random-seed trials, the peak supply-water temperature was 19.2 °C, representing an exceedance of less than 0.2 °C, which is negligible in practical data center environments.

Fig. 15 illustrates the HEX supply water temperature, total cooling system power consumption, and the daily energy-saving ratio over the same 30-day interval. Fig. 15 (a) presents the 30-day temperature regulation performance. The baseline strategy maintains water temperatures below 17 °C approximately 18.5 % of the time and within the 17.0–18.0 °C range for 78.1 % of the time. Although this approach effectively ensures adequate cooling, it results in higher energy consumption. In contrast, the PIML-MPC strategy consistently operates at temperatures above 17.0 °C, with 100 % of the time falling between 18.0 and 19.0 °C. This approach operates closer to the upper threshold without exceeding it, thereby achieving more efficient temperature control. Notably, both methods maintain a zero TVR, indicating strict adherence to thermal constraints. Fig. 15 (b) compares daily power consumption and energy savings for both strategies. Over the 30-day period, PIML-MPC outperforms the baseline every day, achieving energy reductions ranging from 4.6 % to 12.9 %. Additionally, it is worth noting that the 9.1 % energy saving achieved by the PIML-MPC in this instance is lower than the 12.6 % reported in Section 4.1. This discrepancy stems from the baseline's use of constant setpoints tuned for peak loads, which leads to over-cooling and wasted energy under low ITE demand.

4.2.2. Comparison with existing MPC controller

Fig. 16 summarizes the performance of the four MPC variants under previously unseen operating conditions. PIML-MPC achieved the lowest cooling energy consumption (18,047.2 kWh), followed by ANN-MPC (19,045.6 kWh), NARX-MPC (19,750.6 kWh), and ARX-MPC (22,497.5 kWh). In terms of CSE, PIML-MPC attained 0.122, compared with 0.128 for ANN-MPC, 0.132 for NARX-MPC, and 0.150 for ARX-MPC. Regarding temperature regulation, PIML-MPC maintained the supply-water temperature within the prescribed range (mean = 18.6 °C) and achieved a TVR of only 1.0 %, demonstrating both accurate temperature prediction and near-complete constraint compliance. ANN-MPC and NARX-MPC maintained mean supply temperatures of 18.7 °C and 18.9 °C, respectively—both within the acceptable range—but exhibited high temperature violation rates of 24.9 % and 40.5 %, indicating unstable control performance. In contrast, ARX-MPC failed to maintain temperature bounds, with a higher mean temperature of 19.5 °C and a TVR of 57.0 %. In terms of energy consumption, all three benchmark controllers consumed significantly more energy than PIML-MPC. In conclusion, PIML-MPC thus provides the most robust, constraint-compliant operation and outperforms both machine-learning–based and time-series–based MPC approaches under unseen conditions.

It is worth noting that ANN-MPC achieved performance comparable to PIML-MPC under familiar conditions (as shown in Section 4.1.2); however, its performance deteriorated markedly under unseen scenarios, whereas PIML-MPC remained robust. This observation is consistent with the fundamental scientific understanding that machine learning models incorporating physical knowledge possess stronger generalization capability than purely data-driven models.

Fig. 17 presents the HEX supply water temperature and total cooling system power consumption for all four control strategies over a 30-day period (seed 40). It's obvious that all controllers except PIML-MPC exhibited significant temperature oscillations and violations. To quantify these fluctuations, we calculated the daily temperature range—the difference between the highest and lowest temperatures over each 24-h period. ANN-MPC experienced its largest swing of 6.4 °C on day 23; ARX-MPC peaked at 4.5 °C on day 29; and NARX-MPC reached 5.1 °C on day 21. In contrast, PIML-MPC maintained exceptional stability, limiting its daily temperature range to just 0.9 °C. These results demonstrate that PIML-MPC achieves the greatest stability—minimizing both oscillations and violations—by leveraging highly accurate predictions to generate reliable control actions throughout the disturbance period.

Fig. 18 illustrates the performance of three MPC strategies under unseen conditions, evaluated over ten independent trials (random seeds 40–49). Panels (a)–(c) report the temperature-regulation metrics and total cooling energy consumption for each controller, while panels (d)–(f) show the corresponding standard deviations. Notably, PIML-MPC achieves the lowest variability across seeds, with standard deviations of 0.15 °C in the maximum supply water temperature, 0.08 °C in the mean supply water temperature, and 82 kWh in total cooling energy consumption. By contrast, both ANN-MPC and NARX-MPC exhibit larger run-to-run fluctuations. These findings demonstrate that PIML-MPC offers superior robustness to random weight initialization and seed variation, yielding more consistent performance among ML-based controllers.

4.3. Limitations and future work

Despite promising simulation results, several challenges must be addressed before practical deployment, and multiple avenues warrant further investigation. First, all results presented here are derived from a simulation model calibrated with historical measurements from the STDCT data center. While the simulator captures key thermal and hydraulic dynamics, it cannot fully reproduce sensor noise, unmodeled disturbances, or equipment degradation. We will conduct field trials in STDCT to assess real-world efficacy. Inject realistic sensor uncertainty and latency into the control loop to evaluate robustness and compare PIML-MPC against existing industrial controllers under varying ambient and ITE load conditions. Second, once trained, the prediction model parameters remain fixed throughout the control horizon. This assumes that plant dynamics and operating conditions do not drift over time. We plan to develop an adaptive PIML-MPC scheme in which the prediction model is updated online to track slow changes in heat exchanger fouling, pump aging, or thermal load profiles. Explore the trade-off between adaptation speed, data throughput, and controller stability.

5. Conclusions

In this study, we propose a physics-informed, machine learning-based model predictive control framework for intelligent EDC operations. Leveraging operational data from the STDCT, we developed a high-fidelity simulation environment to assess our control strategy.

First, under training conditions, PIML-MPC was benchmarked against a baseline controller. Experimental results demonstrate that PIML-MPC achieves a 0.0 % temperature violation rate while reducing energy consumption by 12.6 %. Next, we evaluated the framework under previously unseen conditions across a wider spectrum of ITE workloads. In these tests, PIML-MPC maintained a mean TVR of 1.0 % and achieved a 9.1 % energy saving compared to the baseline, highlighting its strong generalization capability. Even in the case where violations occurred, the peak supply temperature reached 19.2 °C—just 0.2 °C above the limit, a deviation negligible in practical operation.

Second, we benchmarked PIML-MPC against three prediction-model-based MPC schemes. Under familiar conditions, PIML-MPC achieved the second-lowest energy consumption while maintaining a 0.0 % TVR, demonstrating the best balance between energy efficiency and temperature management. Although ANN-MPC achieved slightly higher energy savings, it failed to regulate temperature effectively. In contrast, ARX-MPC and NARX-MPC consumed more energy and resulted in higher supply-water temperatures. In previously unseen scenarios, all three benchmark controllers exhibited TVRs exceeding 20 %, whereas PIML-MPC consistently outperformed them in both energy efficiency and thermal control. This underscores the critical role of embedded physical information in enhancing the model's generalization capability. Moreover, ten randomized trials further demonstrate that PIML-MPC surpasses purely data-driven and nonlinear time-series-based MPC methods by maintaining minimal performance variability, low sensitivity to initialization. These findings underscore PIML-MPC's suitability for real-world deployment.

In conclusion, our physics-informed, AI-driven MPC framework represents a significant advance in the sustainable, efficient, and reliable management of EDC cooling systems. By harmonizing physical laws with cutting-edge learning techniques, PIML-MPC offers a compelling, ready-for-deployment solution that meets the stringent demands of next-generation edge computing and edge AI infrastructures.

CRediT authorship contribution statement

Dong Chen: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Chee-Kong Chui:** Supervision. **Poh Seng Lee:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Benchmark Models

An ARX model is a linear time-series representation that relates the current output to past outputs and external inputs. It takes the form

$$x^t = \sum_{k=1}^{M} A^k x^{t-k} + \sum_{k=1}^{M} B^k \boldsymbol{u}^{t-k} + Cd^{t-1}, \tag{56}$$

where x^t, $\boldsymbol{u}^k$ and d^t denote the state, control, and disturbance vectors at current time step t; A^k, B^k, and C represent model parameters corresponding to previous time steps indexed by k. M is the model order (here $M = 3$). The parameters $\{A^k, B^k, C\}$ are estimated in closed form by organizing the observed data into a regression matrix and performing a single batch least-squares solution, thereby minimizing the MSE.

A NARX model extends the ARX framework by replacing the linear mapping with a nonlinear function, typically a neural network. The discrete-time predictor takes the form

$$x^t = f_\theta\left(x^{t-1}, \ldots, x^{t-M}, \boldsymbol{u}^{t-1}, \ldots, \boldsymbol{u}^{t-M}, d^{t-1}\right), \tag{57}$$

where $M = 3$ and f_θ is a neural network parameterized by weights θ. These weights are learned via gradient descent by minimizing the mean-squared prediction error.

An ANN model is a fully connected feedforward neural network that maps input features directly to the predicted output. In this study, the input and output features are the same as those used for the APCNN models, as described in Section 2.3. Model parameters are trained using gradient descent to minimize the MSE between predictions and measurements.

Appendix B. Model Training Setup

All neural network variants were implemented in PyTorch. Hyperparameters were tuned using Bayesian Optimization, with up to 100 trials conducted for each model. The selected hyperparameters for the three models are summarized in Table B.1. After hyperparameter selection, the dataset was partitioned into training and validation sets using an 80/20 split. Each model was trained for up to 300 epochs, with early stopping applied if the validation loss did not improve for 10 consecutive epochs, thereby mitigating overfitting and improving training efficiency. Robustness was examined by analyzing control performance across ten random seeds (40–49) for model initialization and data partitioning. Across all neural frameworks, the Rectified Linear Unit (ReLU) activation function was employed, and the Adam optimizer was consistently used for its efficiency and robustness in achieving convergence. The loss function was defined as the MSE augmented by an L2 regularization term weighted by 1×10^{-5}. L2 regularization penalizes large weights, thereby encouraging simpler models that generalize better and reducing overfitting to noisy operational data. It should be noted that the above procedures for hyperparameter optimization, training, and robustness analysis are relevant only to the ANN, NARX, and PIML models. In contrast, the ARX model is a linear regressor whose parameters are obtained in closed form using a least-squares solution. For a fixed dataset and model structure, this yields a unique and deterministic parameter set, rendering stochastic initialization and seed variation irrelevant.

Table B.1
Optimized hyperparameters for three architectures.

Model	ANN	NARX	PIML
Learning rate	2×10^{-4}	1×10^{-4}	5×10^{-4}
Batch size	64	64	64
Neurons per layer	256	256	128
Hidden layers	4	4	4

Appendix C. Computational Burden Analysis

To evaluate the computational feasibility of the proposed framework, we analyze both the training cost of the APCNN and the runtime requirements of MPC solving. The APCNN developed in this study accepts 4–5 input features (depending on the specific component model) and produces a single output variable. Its architecture consists of four hidden layers with 256 neurons each. The training dataset contains 3360 samples with a temporal resolution of 15 min, and the detailed procedure is provided in Appendix B. End-to-end training of the three sub-models on an Intel Core i7-9750H CPU with an NVIDIA GeForce GTX 1650 GPU takes an average of 20.55 s. The average was obtained from ten repeated experiments using different random seeds. The dataset corresponds to roughly one month of operation, and even with multi-year data accumulation, the training cost is expected to remain moderate and manageable.

The MPC problem is formulated with a 15-min control interval and a 30-min prediction horizon, optimizing four decision variables at each step. The ranges and step sizes of these variables are summarized in Table C.1. To ensure computational tractability, a two-stage hybrid optimization strategy is adopted. In the first stage, a coarse global search is performed, in which each variable takes five candidate values centered on its previous action. This design leverages temporal continuity in system dynamics to enhance search efficiency and prevent abrupt control changes. The coarse search thus explores up to $5^4 = 625$ combinations, with fewer when variables reach their boundaries. The three lowest-cost candidates from this search are then retained for subsequent refinement. In the second stage, each of these candidates is refined using the SLSQP solver, configured with a maximum of 100 iterations. These settings ensure reliable convergence while preserving computational efficiency. On the specified hardware, the average optimization runtime per control step is approximately 2.73 s, based on ten repeated experiments with different random seeds. This runtime is significantly lower than the 15-min control interval, confirming the feasibility of real-time MPC deployment.

Overall, both APCNN training and MPC solving impose modest computational demands, demonstrating the practicality of the proposed framework for real-time implementation in edge data centers.

Table C.1
Control variable ranges and step sizes used in MPC.

Actions	Minimum	Maximum	Step interval
Chiller supply water temperature	12.0	18.0	0.1
Chilled water pump setpoint	0.3	1.0	0.01
Condensed water pump setpoint	0.3	1.0	0.01
Cooling tower fan speed	0.3	1.0	0.01

Data availability

Data will be made available on request.

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